



Sparse-grid quadrature nonlinear filtering[☆]

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ABSTRACT

In this paper, a novel nonlinear filter named Sparse-grid Quadrature Filter (SGQF) is proposed. The filter utilizes weighted sparse-grid quadrature points to approximate the multi-dimensional integrals in the nonlinear Bayesian estimation algorithm. The locations and weights of the univariate quadrature points with a range of accuracy levels are determined by the moment matching method. Then the univariate quadrature point sets are extended to form a multi-dimensional grid using the sparse-grid theory. Compared with the conventional point-based methods, the estimation accuracy level of the SGQF can be flexibly controlled and the number of sparse-grid quadrature points for the SGQF is a polynomial of the dimension of the system, which alleviates the curse of dimensionality for high dimensional problems. The Unscented Kalman Filter (UKF) is proven to be a subset of the SGQF at the level-2 accuracy. The performance of this filter is demonstrated by an orbit estimation problem. The simulation results show that the SGQF achieves higher accuracy than the Extended Kalman Filter (EKF), the UKF, and the Cubature Kalman Filter (CKF). In addition, the SGQF is computationally much more efficient than the multi-dimensional Gauss–Hermite Quadrature Filter (GHQF) with the same performance.

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1. Introduction

Nonlinear filtering has been widely used in many science and engineering disciplines. The general filtering problem can be addressed using the Bayesian estimation theory (Jazwinski, 1970). In the Bayesian framework, the probability density function (pdf) of the states of a continuous dynamic system is propagated between two consecutive measurements by solving the Fokker–Planck–Kolmogorov equation (FPKE) (Challa & Bar-Shalom, 2000; Daum, 2005) and the posterior density of the states is updated by the Bayesian formula. For discrete dynamic systems, the FPKE will be replaced by the Chapman–Kolmogorov equation. In general, exact finite-dimensional solutions to the integral equations in the Bayesian estimation do not exist. Numerous approximate nonlinear filters have been proposed, of which a large class is based on the assumption that the state pdf is Gaussian. The Extended Kalman Filter (EKF) (Simon, 2006), by far the most widely used nonlinear filter, belongs to this class. The EKF is based on the first-order linearization of the nonlinear models. But the first-order

linearization may fail when the nonlinear system contains large initial estimation errors, large process noise, or high nonlinearities. In Crassidis and Markley (2003), a practical spacecraft attitude estimation problem is used to show that when the initial estimation errors (attitude errors and gyro bias errors) are large, the EKF fails whereas the Unscented Kalman Filter (UKF) maintains acceptable accuracy.

There has been intensive research on new nonlinear filtering methods to improve the performance of the EKF. A large class of such nonlinear filters that has drawn a lot of attention in recent decades are the point-based nonlinear filters. These filters can capture higher-order moments more accurately by using selected points to approximate the pdf according to certain rules. They include the Gauss–Hermite Quadrature Filter (GHQF) based on the Gauss–Hermite Quadrature (GHQ) rule (Arasaratnam, Haykin, & Elliott, 2007; Ito & Xiong, 2000), the UKF based on the Unscented Transformation (UT) (Julier & Uhlmann, 2004; Julier, Uhlmann, & Durrant-Whyte, 2000), the Cubature Kalman Filter (CKF) based on the Cubature rule (Arasaratnam & Haykin, 2009; Arasaratnam, Haykin, & Hurd, 2010), the Central Difference Filter (CDF) (Ito & Xiong, 2000), the Divided Difference Filter (DDF) (Nørgaard, Poulsen, & Ravn, 2000), and the particle filter (Arulampalam, Maskell, Gordon, & Clapp, 2002; Doucet, Godsill, & Andrieu, 2000; Ristic, Arulampalam, & Gordon, 2004). Compared with the EKF, the point-based methods have two important advantages: (1) derivative free and (2) achieving higher order approximation to nonlinear functions. All the Gaussian approximation filters involve using numerical quadratures to approximate the expectations of nonlinear functions with respect to Gaussian pdfs. The GHQF

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evaluates the nonlinear functions at the Gaussian quadrature points, whose number grows exponentially with the state dimension. The UKF uses the so-called sigma points with the number of points increasing linearly with the dimension. The different point-selection strategies of the UKF include the minimal-skew simplex points (Julier & Uhlmann, 2002), the spherical simplex points (Julier, 2003; Lozana, Carrillo, Dzul, & Lozana, 2008), the Schmidt orthogonalization-based simplex set (Li, Wei, & Xiao, 2007), and those with more than $2n + 1$ sigma points for accuracy improvement (Lévesque, 2006; Sun, Ma, & Wang, 2009; Van Zandt, 2001), where n is the dimension. The CDF and DDF are based on interpolation formula using the similar deterministic sampling approach. For n -dimensional systems, the original CKF and UKF using symmetric $2n$ and $2n + 1$ points respectively are accurate up to the 3rd-order polynomials. The simplex UKF and the spherical simplex UKF can be partly accurate to the 3rd-order polynomials using $n + 2$ nonsymmetrical points. The accuracy of different UKF methods has been compared in Zhang, Liu, and Zhao (2009). The common limitation of those methods including the UKF, CDF, DDF and CKF is that they are hard to extend to achieve any higher-order estimation accuracy.

The point based methods are also used for non-Gaussian systems by the Gaussian mixture method (Arasaratnam et al., 2007; Faubel, McDonough, & Klakow, 2009) or the particle filter (Van der Merwe, Doucet, De Freitas, & Wan, 2000). The particle filter (Arulampalam et al., 2002) can achieve any accuracy levels theoretically, but the computational load is usually prohibitive even for moderately high dimensional dynamical systems.

In this paper, we propose a novel point-based nonlinear filtering algorithm, the Sparse-grid Quadrature Filter (SGQF), which combines the general moment matching method for one-dimensional integral approximation with the sparse-grid theory for multi-dimensional extension. The terminology sparse-grid filtering has been mentioned in Balaji and Damini (2008). However, the method used therein is based on the sparse tensor technology that is different from the proposed method in this paper. The sparse-grid method was originally used to alleviate the curse of dimensionality problem in numerical integration (Heiss & Winschel, 2008; Smolyak, 1963). The Smolyak rule (Smolyak, 1963) is used in this paper to extend the one-dimensional quadrature point set to the multi-dimensional point set by virtue of a special linear combination of lower-level tensor products such that the number of necessary points is significantly less than that of using the direct tensor product rule. As a result, the computational cost of the sparse-grid method does not increase exponentially.

The SGQF proposed in this paper is more flexible to use than the UKF, CKF, CDF, and DDF in terms of the point selection strategies, and capable of achieving higher accuracy levels with only moderately increased number of points, which is more efficient than the GHQF and particle filters. Another important contribution of this paper is to prove that the UKF is a subset of the SGQF at the level-2 accuracy.

The rest of this paper is organized as follows: the Gaussian approximation filters are briefly reviewed in Section 2. Section 3 introduces the sparse-grid quadrature (SGQ) rule and the SGQF as well as the theoretical results. Section 4 gives the simulation results illustrating the performance of the SGQF for the orbit estimation problem and compares the SGQF with the EKF, GHQF, CKF, and UKF. Concluding remarks are given in Section 5.

2. Gaussian approximation filters

In this section, the nonlinear filters based on the Gaussian approximation to the Bayesian estimation framework are briefly reviewed.

2.1. Bayesian nonlinear filtering framework

Consider a nonlinear discrete-time system given by

$$\mathbf{x}_k = \mathbf{f}(\mathbf{x}_{k-1}) + \mathbf{v}_{k-1} \quad (1)$$

$$\mathbf{y}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{n}_k \quad (2)$$

where $\mathbf{x}_k$ is the $n \times 1$ state vector and $\mathbf{y}_k$ is the $m \times 1$ measurement vector; $\mathbf{v}_{k-1}$ and $\mathbf{n}_k$ are independent white Gaussian process noise and measurement noise with the covariance $\mathbf{Q}_{k-1}$ and $\mathbf{R}_k$ respectively. This additive noise model is commonly used in the point-based filters.

The Bayesian filtering includes recursive prediction and update procedures (Jazwinski, 1970). Given the prior pdf of the system $p(\mathbf{x}_{k-1}|\mathbf{y}_{1:k-1})$, the conditional pdf $p(\mathbf{x}_k|\mathbf{y}_{1:k-1})$ satisfies the Chapman–Kolmogorov equation (Jazwinski, 1970).

$$p(\mathbf{x}_k|\mathbf{y}_{1:k-1}) = \int p(\mathbf{x}_k|\mathbf{x}_{k-1}) p(\mathbf{x}_{k-1}|\mathbf{y}_{1:k-1}) d\mathbf{x}_{k-1}. \quad (3)$$

When the measurement at time k is available, the posterior conditional pdf satisfies the following equation:

$$p(\mathbf{x}_k|\mathbf{y}_{1:k}) = \frac{p(\mathbf{y}_k|\mathbf{x}_k) p(\mathbf{x}_k|\mathbf{y}_{1:k-1})}{\int p(\mathbf{y}_k|\mathbf{x}_k) p(\mathbf{x}_k|\mathbf{y}_{1:k-1}) d\mathbf{x}_k} \quad (4)$$

where $p(\mathbf{y}_k|\mathbf{x}_k)$ is the likelihood function.

Equation (3) is called the prediction formula while Eq. (4) is called the update formula. The moments, such as mean and covariance, can be calculated from the conditional pdf $p(\mathbf{x}_k|\mathbf{y}_{1:k})$.

The recursive propagation of the posterior density given by Eqs. (3) and (4) is only a conceptual solution in the sense that it cannot be determined analytically in general. Thus, one has to use approximate or suboptimal Bayesian algorithms.

2.2. Gaussian approximation filters

Assuming that the pdf of the states is Gaussian, the Gaussian approximation filters can be derived from the Bayesian estimation Eqs. (3) and (4) (Arasaratnam et al., 2007; Ito & Xiong, 2000; Julier & Uhlmann, 2004). If the noise is non-additive, a more general framework can be used as in Julier and Uhlmann (2004). With this Gaussian assumption, the filter algorithm can be conveniently designed and the filter structure is relatively simple because only the mean and covariance need to be calculated. The Gaussian approximation filters have been shown in practice to be effective in many applications including nonlinear and non-Gaussian systems.

Prediction:

$$\hat{\mathbf{x}}_{k|k-1} = \int_{\mathbf{R}^n} \mathbf{f}(\mathbf{x}_{k-1}) N(\mathbf{x}_{k-1}; \hat{\mathbf{x}}_{k-1|k-1}, \mathbf{P}_{k-1|k-1}) d\mathbf{x}_{k-1} \quad (5)$$

$$\mathbf{P}_{k|k-1} = \int_{\mathbf{R}^n} \mathbf{f}(\mathbf{x}_{k-1}) \mathbf{f}^T(\mathbf{x}_{k-1}) N(\mathbf{x}_{k-1}; \hat{\mathbf{x}}_{k-1|k-1}, \mathbf{P}_{k-1|k-1}) d\mathbf{x}_{k-1} - \hat{\mathbf{x}}_{k|k-1} \hat{\mathbf{x}}_{k|k-1}^T + \mathbf{Q}_{k-1} \quad (6)$$

where $N(\mathbf{x}_{k-1}; \hat{\mathbf{x}}_{k-1|k-1}, \mathbf{P}_{k-1|k-1})$ denotes the multivariate normal distribution with the mean $\hat{\mathbf{x}}_{k-1|k-1}$ and the covariance $\mathbf{P}_{k-1|k-1}$.

Update:

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{L}_k (\mathbf{y}_k - \mathbf{z}_k) \quad (7)$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{L}_k \mathbf{P}_{xz}^T \quad (8)$$

where

$$\mathbf{L}_k = \mathbf{P}_{xz} (\mathbf{R}_k + \mathbf{P}_{zz})^{-1} \quad (9)$$

$$\mathbf{z}_k = \int_{\mathbf{R}^n} \mathbf{h}(\mathbf{x}_k) N(\mathbf{x}_k; \hat{\mathbf{x}}_{k|k-1}, \mathbf{P}_{k|k-1}) d\mathbf{x}_k \quad (10)$$

$$\mathbf{P}_{xz} = \int_{\mathbf{R}^n} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}) (\mathbf{h}(\mathbf{x}_k) - \mathbf{z}_k)^T \times N(\mathbf{x}_k; \hat{\mathbf{x}}_{k|k-1}, \mathbf{P}_{k|k-1}) d\mathbf{x}_k \quad (11)$$

$$\mathbf{P}_{zz} = \int_{\mathbf{R}^n} (\mathbf{h}(\mathbf{x}_k) - \mathbf{z}_k) (\mathbf{h}(\mathbf{x}_k) - \mathbf{z}_k)^T \times N(\mathbf{x}_k; \hat{\mathbf{x}}_{k|k-1}, \mathbf{P}_{k|k-1}) d\mathbf{x}_k. \quad (12)$$

Since the integrals in Eqs. (5), (6) and (10)–(12) follow the special form $\int_{\mathbf{R}^n} \mathbf{F}(\mathbf{x}) N(\mathbf{x}; \hat{\mathbf{x}}, \mathbf{P}) d\mathbf{x}$, where $\mathbf{F}(\mathbf{x})$ is a general nonlinear function representing the functions in Eqs. (5), (6) and (10)–(12), they can be approximated by many numerical rules, such as, the GHQ rule, the UT, or the Cubature rule. The general quadrature rule with respect to the Gaussian distribution with zero mean and unit covariance can be approximated by

$$\int_{\mathbf{R}^n} \mathbf{F}(\mathbf{x}) N(\mathbf{x}; \mathbf{0}, \mathbf{I}) d\mathbf{x} \approx \sum_{i=1}^{N_p} W_i \mathbf{F}(\boldsymbol{\gamma}_i) \quad (13)$$

where N_p is the total number of points; $\boldsymbol{\gamma}_i$ and W_i are the quadrature points and weights respectively.

The integrals with respect to a more general Gaussian distribution $\int_{\mathbf{R}^n} \mathbf{F}(\mathbf{x}) N(\mathbf{x}; \hat{\mathbf{x}}, \mathbf{P}) d\mathbf{x}$ can be approximated by Arasaratnam et al. (2007) Eq. (14) using covariance transformation:

$$\begin{aligned} \int_{\mathbf{R}^n} \mathbf{F}(\mathbf{x}) N(\mathbf{x}; \hat{\mathbf{x}}, \mathbf{P}) d\mathbf{x} &= \int_{\mathbf{R}^n} \mathbf{F}(\mathbf{S}\boldsymbol{\gamma} + \hat{\mathbf{x}}) N(\boldsymbol{\gamma}; \mathbf{0}, \mathbf{I}) d\boldsymbol{\gamma} \\ &\approx \sum_{i=1}^{N_p} W_i \mathbf{F}(\mathbf{S}\boldsymbol{\gamma}_i + \hat{\mathbf{x}}) \end{aligned} \quad (14)$$

where $\mathbf{P} = \mathbf{S}\mathbf{S}^T$.

Applying Eq. (14) to approximate the integrals in Eqs. (5), (6) and (10)–(12), the filtering algorithm can be rewritten as follows.

Prediction:

$$\hat{\mathbf{x}}_{k|k-1} = \sum_{i=1}^{N_p} W_i \mathbf{f}(\boldsymbol{\xi}_i) \quad (15)$$

$$\mathbf{P}_{k|k-1} = \sum_{i=1}^{N_p} W_i (\mathbf{f}(\boldsymbol{\xi}_i) - \hat{\mathbf{x}}_{k|k-1}) (\mathbf{f}(\boldsymbol{\xi}_i) - \hat{\mathbf{x}}_{k|k-1})^T + \mathbf{Q}_{k-1} \quad (16)$$

where $\boldsymbol{\xi}_i$ is the transformed point obtained from the covariance decomposition, i.e.

$$\mathbf{P}_{k-1|k-1} = \mathbf{S}\mathbf{S}^T; \quad \boldsymbol{\xi}_i = \mathbf{S}\boldsymbol{\gamma}_i + \hat{\mathbf{x}}_{k-1|k-1}. \quad (17)$$

Update:

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{L}_k (\mathbf{y}_k - \mathbf{z}_k) \quad (18)$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{L}_k \mathbf{P}_{xz}^T \quad (19)$$

where $\mathbf{L}_k$ is calculated from (9);

$$\mathbf{z}_k = \sum_{i=1}^{N_p} W_i \mathbf{h}(\tilde{\boldsymbol{\xi}}_i) \quad (20)$$

$$\mathbf{P}_{xz} = \sum_{i=1}^{N_p} W_i (\tilde{\boldsymbol{\xi}}_i - \hat{\mathbf{x}}_{k|k-1}) (\mathbf{h}(\tilde{\boldsymbol{\xi}}_i) - \mathbf{z}_k)^T \quad (21)$$

$$\mathbf{P}_{zz} = \sum_{i=1}^{N_p} W_i (\mathbf{h}(\tilde{\boldsymbol{\xi}}_i) - \mathbf{z}_k) (\mathbf{h}(\tilde{\boldsymbol{\xi}}_i) - \mathbf{z}_k)^T \quad (22)$$

$\tilde{\boldsymbol{\xi}}_i$ is the transformed point obtained from the predicted covariance decomposition, i.e.

$$\mathbf{P}_{k|k-1} = \tilde{\mathbf{S}}\tilde{\mathbf{S}}^T; \quad \tilde{\boldsymbol{\xi}}_i = \tilde{\mathbf{S}}\boldsymbol{\gamma}_i + \hat{\mathbf{x}}_{k|k-1}. \quad (23)$$

The quadrature points $\boldsymbol{\gamma}_i$ and the weights W_i can be chosen according to the GHQ rule (Arasaratnam et al., 2007; Ito & Xiong, 2000), UT (Julier & Uhlmann, 2004), or Cubature rule (Arasaratnam & Haykin, 2009).

2.2.1. Gauss–Hermite quadrature rule

For the univariate GHQ rule with respect to the univariate standard Gaussian distribution $N(x; 0, 1)$ with m_q quadrature points, the univariate point γ_i and its weight w_i can be calculated as follows. If $m_q = 1$, then $\gamma_1 = 0$ and $w_1 = 1$. If $m_q > 1$, first construct a symmetric tri-diagonal matrix J with zero diagonal elements and $J_{i,i+1} = J_{i+1,i} = \sqrt{i/2}$, $1 \leq i \leq m_q - 1$. Then the i th quadrature point γ_i is calculated by $\gamma_i = \sqrt{2}\varepsilon_i$, where ε_i is the i th eigenvalue of J . The corresponding weight w_i is calculated by $w_i = (\mathbf{v}_i)_1^2$ where $(\mathbf{v}_i)_1$ is the first element of the i th normalized eigenvector of J . The univariate GHQ rule with m_q quadrature points is exact up to the $(2m_q - 1)$ th order of polynomials (Arasaratnam et al., 2007). The multivariate GHQ rule extends the above univariate GHQ rule by the direct tensor product (Arasaratnam et al., 2007; Ito & Xiong, 2000). For example, the two dimensional GHQ points $\boldsymbol{\gamma}_i$ and weights W_i constructed from 3 univariate GHQ points $\gamma_1, \gamma_2, \gamma_3$ with corresponding weights w_1, w_2, w_3 by the tensor product rule contains $3^2 = 9$ points

$$\begin{pmatrix} (\gamma_1, \gamma_1), (\gamma_1, \gamma_2), (\gamma_1, \gamma_3) \\ (\gamma_2, \gamma_1), (\gamma_2, \gamma_2), (\gamma_2, \gamma_3) \\ (\gamma_3, \gamma_1), (\gamma_3, \gamma_2), (\gamma_3, \gamma_3) \end{pmatrix}$$

and 9 corresponding weights

$$\begin{pmatrix} w_1 w_1, w_1 w_2, w_1 w_3 \\ w_2 w_1, w_2 w_2, w_2 w_3 \\ w_3 w_1, w_3 w_2, w_3 w_3 \end{pmatrix}$$

where $\gamma_1 = -\sqrt{3}$, $\gamma_2 = 0$, and $\gamma_3 = \sqrt{3}$; $w_1 = 1/6$, $w_2 = 2/3$, and $w_3 = 1/6$.

The multivariate GHQ rule is exact for all polynomials of the form $x_1^{i_1} x_2^{i_2} \dots x_n^{i_n}$ with $1 \leq i_j \leq 2m_q - 1$ (Ito & Xiong, 2000). However, the total number of points $N_p = (m_q)^n$ increases exponentially with the dimension n , which is the curse of dimensionality problem.

2.2.2. Unscented transformation

For the UT with $2n + 1$ sigma points (Julier et al., 2000), $\boldsymbol{\gamma}_i$ and W_i are given by

$$\begin{cases} \boldsymbol{\gamma}_1 = [0, 0, \dots, 0, 0]^T; & W_1 = \frac{\kappa}{n + \kappa}; & i = 1 \\ \boldsymbol{\gamma}_i = \sqrt{n + \kappa} \mathbf{e}_{i-1}; & W_i = \frac{1}{2(n + \kappa)}; & 2 \leq i \leq n + 1 \\ \boldsymbol{\gamma}_i = -\sqrt{n + \kappa} \mathbf{e}_{i-n-1}; & W_i = \frac{1}{2(n + \kappa)}; & n + 2 \leq i \leq 2n + 1 \end{cases} \quad (24)$$

where $\mathbf{e}_{i-1}$ is the unit vector in $\mathbf{R}^n$ with the $(i-1)$ th element being 1 and κ is a tunable parameter of which the suggested optimal value for Gaussian distributions is $\kappa = 3 - n$ (Julier et al., 2000).

2.2.3. Cubature rule

Recently, a new numerical rule named cubature rule has been proposed to calculate the integrals. The points $\mathbf{y}_i$ and weights W_i of the cubature rule are given in Arasaratnam and Haykin (2009).

$$\begin{cases} \mathbf{y}_i = \sqrt{n} \mathbf{e}_{i-1}; & W_i = \frac{1}{2n}; & 1 \leq i \leq n \\ \mathbf{y}_i = -\sqrt{n} \mathbf{e}_{i-n-1}; & W_i = \frac{1}{2n}; & n+1 \leq i \leq 2n. \end{cases} \quad (25)$$

It can be verified that Eq. (25) is identical to Eq. (24) when $\kappa = 0$. Although the cubature rule and UT are identical when $\kappa = 0$, they are based on different numerical rules.

In this paper, we propose a new point selection strategy that can include the sigma points as a subset and achieve higher estimation accuracy. In addition, it requires a significantly fewer points based on the sparse-grid method than the multivariate GHQ rule.

3. Sparse-grid quadrature filter (SGQF)

In this section, we introduce a new multi-dimensional quadrature rule that is based on the moment matching method and the sparse-grid theory. The structure of the SGQF algorithm is the same as that of other Gaussian approximation filters, i.e., Eqs. (15)–(23).

3.1. Sparse-grid quadrature rule

In this section, the sparse-grid quadrature (SGQ) is proposed to approximate the integrals in Eqs. (5)–(6) and (10)–(12) based on the sparse-grid method. The sparse-grid method utilizes a linear combination of lower-level tensor products of univariate quadrature rules to approximate multivariate integrals. We will refer to the points to do this approximation as the SGQ points. The sparse-grid method is based on the Smolyak rule, given by Heiss and Winschel (2008)

$$\begin{aligned} \int_{\mathbf{R}^n} \mathbf{f}(\mathbf{x}) N(\mathbf{x}; \mathbf{0}, \mathbf{I}_n) d\mathbf{x} &\approx I_{n,L}(\mathbf{f}) \\ &= \sum_{q=L-n}^{L-1} (-1)^{L-1-q} C_{n-1}^{L-1-q} \sum_{\mathcal{E} \in \mathbf{N}_q^n} (I_{i_1} \otimes \cdots \otimes I_{i_n})(\mathbf{f}) \end{aligned} \quad (26)$$

where $\mathbf{x} = [x_1, x_2, \dots, x_p, \dots, x_n]^T \in \mathbf{R}^n$; $I_{n,L}(\mathbf{f})$ is an approximation to the n -dimensional integral of the function f with respect to $N(\mathbf{x}; \mathbf{0}, \mathbf{I}_n)$ with the accuracy level $L \in \mathbb{N}$, where $\mathbb{N}$ is the set of natural numbers. By accuracy level L , it means that $I_{n,L}(\mathbf{f})$ is exact for all polynomials of the form $x_1^{i_1} x_2^{i_2} \cdots x_n^{i_n}$ with $\sum_{j=1}^n i_j \leq 2L-1$ (Heiss & Winschel, 2008). C_{n-1}^{L-1-q} denotes the binomial coefficient, defined by $C_n^k = n!/[k!(n-k)!]$, where $n!$ denotes the factorial of n ; $\otimes$ denotes the tensor product. I_{i_j} is the univariate quadrature rule with the accuracy level of $i_j \in \mathcal{E}$, where $\mathcal{E} \triangleq (i_1, \dots, i_n)$ is an accuracy level sequence of n natural numbers. By accuracy level i_j , it means that I_{i_j} is exact to at least the $(2i_j - 1)$ th order of all univariate polynomials. $\mathbf{N}_q^n$ is the set of accuracy level sequences defined by

$$\begin{cases} \mathbf{N}_q^n = \left\{ \mathcal{E} : \sum_{j=1}^n i_j = n+q \right\} & \text{for } q \geq 0 \\ \mathbf{N}_q^n = \emptyset & \text{for } q < 0 \end{cases} \quad (27)$$

where q is an auxiliary parameter that has the range of $L-n \leq q \leq L-1$, and $\emptyset$ is the empty set.

Note: $\mathbf{N}_q^n$ determines the possible accuracy level i_j for each univariate quadrature and q prescribes the range of the accuracy level i_j in each dimension.

The more explicit form of Eq. (26) can be written as

$$\begin{aligned} I_{n,L}(\mathbf{f}) &= \sum_{q=L-n}^{L-1} \sum_{\mathcal{E} \in \mathbf{N}_q^n} \sum_{\bar{p}_{s_1} \in X_{i_1}} \cdots \sum_{\bar{p}_{s_n} \in X_{i_n}} \mathbf{f}(\bar{p}_{s_1}, \dots, \bar{p}_{s_n}) \\ &\times \left\{ (-1)^{L-1-q} C_{n-1}^{L-1-q} \prod_{j=1}^n \bar{w}_{s_j} \right\} \end{aligned} \quad (28)$$

where X_{i_j} is the univariate quadrature point set of the univariate quadrature rule I_{i_j} with the accuracy level i_j , which contains i_j or more points. The choice of X_{i_j} is not unique. The moment matching method will be used in this paper to select the univariate quadrature points in X_{i_j} and will be discussed later.

$[\bar{p}_{s_1}, \dots, \bar{p}_{s_j}, \dots, \bar{p}_{s_n}]^T$ is an n -dimensional SGQ point and $\bar{p}_{s_j} \in X_{i_j}$; $\bar{w}_{s_j}$ is the weight in I_{i_j} associated with the point $\bar{p}_{s_j}$ and the term enclosed by the braces is the weight associated with an n -dimensional SGQ point determined by the accuracy level sequence in $\mathbf{N}_q^n$. For a SGQ point that appears multiple times, the final weight of this point is the sum of the weights on the point over all combinations of $X_{i_1} \otimes X_{i_2} \otimes \cdots \otimes X_{i_n}$ containing that point, which will be shown in Algorithm 1.

The set of (distinct) SGQ points $X_{n,L}$ is given by

$$X_{n,L} = \bigcup_{q=L-n}^{L-1} \bigcup_{\mathcal{E} \in \mathbf{N}_q^n} (X_{i_1} \otimes \cdots \otimes X_{i_n}) \quad (29)$$

where $\bigcup$ denotes the union of the quadrature point sets. Each element in $\mathbf{N}_q^n$, i.e. $\mathcal{E} = (i_1, \dots, i_n)$, determines a tensor product of X_{i_j} where $i_j \in \mathcal{E} \in \mathbf{N}_q^n$. It can be shown later in Proposition 3.1 that the number of SGQ points increases polynomially with the dimension n .

The SGQ method is based on the following theorem.

Theorem 3.1 (Heiss and Winschel (2008)). Assume that the sequence of univariate quadrature rules $I = \{I_l : l \in \mathbb{N}\}$ is defined such that I_l is exact for all univariate polynomials of the order up to $2l-1$. Then the Smolyak rule $I_{n,L}$ (26) using I as the univariate basis sequence is exact for n -variate polynomials of the total order up to $2L-1$.

Note that according to Eqs. (26) and (27), the accuracy level of $i_j \in \mathcal{E}$ for the univariate quadrature rule satisfies $1 \leq i_j \leq L$.

Using Theorem 3.1, to generate an n -dimensional sparse grid with accuracy level L , the following two steps are needed:

Step 1: choose the univariate quadrature point sets and weights with accuracy levels from 1 to L using the moment matching method.

Step 2: extend the univariate quadrature point sets and weights to the n -dimensional SGQ points and weights using the sparse-grid method, i.e. Eqs. (28) and (29).

These two steps are described in the following two sections.

3.2. Univariate quadrature points and weights selection with the moment matching method

For the n -dimensional SGQ with the accuracy level L to be exact for all multivariate polynomials with the total order of $2L-1$, the univariate quadrature rule I_{i_j} ($i_j \in \mathcal{E}$ and $1 \leq i_j \leq L$) needs to be exact for all univariate polynomials of the order up to $2i_j-1$, which can be satisfied using the following moment matching method.

The general moment matching formula used here (one-dimensional Gaussian type integral) is

$$M_j = \int_{-\infty}^{\infty} x^j N(x; 0, 1) dx = \sum_{i=1}^{N_u} \bar{w}_i \bar{p}_i^j \quad (30)$$

where M_j is the j th order moment; N_u is the number of univariate quadrature points; $\bar{p}_i$ and $\bar{w}_i$ are the univariate quadrature points and weights, respectively. We require that the quadrature rule be exact for all univariate polynomials of the order up to $N_u - 1$. Hence, $\bar{p}_i$ and $\bar{w}_i$ should satisfy the following equation.

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ \bar{p}_1 & \bar{p}_2 & \cdots & \bar{p}_{N_u} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{p}_1^{N_u-1} & \bar{p}_2^{N_u-1} & \cdots & \bar{p}_{N_u}^{N_u-1} \end{bmatrix} \begin{bmatrix} \bar{w}_1 \\ \bar{w}_2 \\ \vdots \\ \bar{w}_{N_u} \end{bmatrix} = \begin{bmatrix} M_0 \\ M_1 \\ \vdots \\ M_{N_u-1} \end{bmatrix}. \quad (31)$$

Note that normalization constraint exists since $\sum_{i=1}^{N_u} \bar{w}_i = M_0 = 1$.

The values of $\bar{p}_i$ and $\bar{w}_i$ ($i = 1 \cdots N_u$) can be obtained by solving the nonlinear equations (31). But the solution is not unique in general. Note that when we choose distinct points $\bar{p}_i$ ($i = 1 \cdots N_u$), the coefficient matrix is the Vandermonde matrix whose determinant is nonzero and the weights $\bar{w}_i$ ($i = 1 \cdots N_u$) can be uniquely determined. The univariate GHQ rule is obtained when $\bar{p}_i$ are chosen to be the roots of Hermite polynomials (Arasaratnam et al., 2007; Press, Teukolsky, Vetterling, & Flannery, 1992). Because the N_u -point univariate GHQ is exact for polynomials of the order up to $(2N_u - 1)$ as mentioned in Section 2.2.1, the GHQ points can match up to the $(2N_u - 1)$ th moments.

In this paper, we use $N_u = 2l - 1$ symmetric points about the origin for the univariate quadrature point set with the accuracy level l . The locations of the points are treated as free parameters; the weights are uniquely determined from the point locations. Other point selection methods are also possible, which will be discussed afterward. Because symmetric points are used, we use different notations from those in Eq. (31) to represent the points ($\tilde{p}_i$) and weights ($\tilde{w}_i$). We also assume that the weights for $\tilde{p}_i$ and $-\tilde{p}_i$ are both $\tilde{w}_i$. Using this point-selection strategy, Eqs. (30) and (31) can be rewritten as Eqs. (32) and (33), respectively.

$$M_j = \begin{cases} \tilde{w}_1 + \sum_{i=2}^{(N_u-1)/2+1} 2\tilde{w}_i & j = 0 \\ 0 & j \text{ is odd} \\ \tilde{w}_1 \tilde{p}_1^j + \sum_{i=2}^{(N_u-1)/2+1} 2\tilde{w}_i \tilde{p}_i^j & j \text{ is even.} \end{cases} \quad (32)$$

Because $N_u = 2l - 1$, we have

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0^2 & \tilde{p}_2^2 & \cdots & \tilde{p}_l^2 \\ \vdots & \vdots & \ddots & \vdots \\ 0^{2l-2} & \tilde{p}_2^{2l-2} & \cdots & \tilde{p}_l^{2l-2} \end{bmatrix} \begin{bmatrix} \tilde{w}_1 \\ 2\tilde{w}_2 \\ \vdots \\ 2\tilde{w}_l \end{bmatrix} = \begin{bmatrix} M_0 \\ M_2 \\ \vdots \\ M_{2l-2} \end{bmatrix}. \quad (33)$$

In Eq. (33), only the even moments are considered because the symmetry of the points implies that the odd moments are matched automatically. If we choose $2l - 1$ symmetric and distinct points, the coefficient matrix of Eq. (33) is a Vandermonde matrix. The weights $\tilde{w}_i$ ($i = 1, \dots, l$) can be uniquely determined since the inverse of the Vandermonde matrix has an analytical form (Macon & Spitzbart, 1958).

In the following, we use the SGQ with $L = 3$ as an example to illustrate the moment matching method. According to the Smolyak rule Eqs. (26), or (27), we need the univariate quadrature rules at level-1 (I_1), level-2 (I_2), and level-3 (I_3). For the univariate quadrature rule I_1 , the point set is chosen to be $\{0\}$ with the corresponding weight of 1. For the univariate quadrature rule I_2 , we choose the symmetric univariate quadrature point set as $\{-\hat{p}_1, 0, \hat{p}_1\}$ with the corresponding weight sequence $(\hat{w}_2, \hat{w}_1, \hat{w}_2)$. For the univariate quadrature rule I_3 , we choose the symmetric univariate quadrature point set as $\{-\hat{p}_3, -\hat{p}_2, 0, \hat{p}_2, \hat{p}_3\}$ with the corresponding weight sequence $(\hat{w}_5, \hat{w}_4, \hat{w}_3, \hat{w}_4, \hat{w}_5)$. To satisfy the condition of

Theorem 3.1, these univariate point sets for level-2 and level-3 should match univariate polynomials up to the 3rd order and the 5th order respectively.

Note that the above set of notations for the univariate quadrature points and weights is different from the notations used in the general univariate moment matching Eqs. (30)–(33) because the SGQ involves different levels of univariate quadrature points and weights and is difficult to adopt a uniform set of notations for all levels. From Eq. (33), for level-2, $\hat{p}_2 = \hat{p}_1, \hat{w}_1 = \hat{w}_1$, and $\hat{w}_2 = \hat{w}_2$, the following equations should be satisfied.

$$\begin{cases} \hat{w}_1 + 2\hat{w}_2 = M_0 = 1 \\ 2\hat{w}_2 \hat{p}_1^2 = M_2 = 1. \end{cases} \quad (34)$$

Solving these two equations for $\hat{w}_1$ and $\hat{w}_2$ leads to

$$\begin{cases} \hat{w}_1 = 1 - 1/\hat{p}_1^2 \\ \hat{w}_2 = 1/(2\hat{p}_1^2). \end{cases} \quad (35)$$

If the point set $\{-\hat{p}_1, 0, \hat{p}_1\}$ is the set of the univariate GHQ points, $\hat{p}_1 = \sqrt{3}$, $\hat{w}_1 = \frac{2}{3}$, $\hat{w}_2 = \frac{1}{6}$ (see Section 2.2.1).

Similarly, for level-3, $\hat{p}_2 = \hat{p}_3, \hat{p}_3 = \hat{p}_3, \hat{w}_1 = \hat{w}_3, \hat{w}_2 = \hat{w}_4$ and $\hat{w}_3 = \hat{w}_5$, the following equations should be satisfied

$$\begin{cases} \hat{w}_3 + 2\hat{w}_4 + 2\hat{w}_5 = M_0 = 1 \\ 2\hat{w}_4 \hat{p}_2^2 + 2\hat{w}_5 \hat{p}_3^2 = M_2 = 1 \\ 2\hat{w}_4 \hat{p}_2^4 + 2\hat{w}_5 \hat{p}_3^4 = M_4 = 3. \end{cases} \quad (36)$$

Solving these equations (if $\hat{p}_3 \neq \hat{p}_2$) for $\hat{w}_3, \hat{w}_4, \hat{w}_5$ yields

$$\begin{cases} \hat{w}_3 = 1 - 2\hat{w}_4 - 2\hat{w}_5 \\ \hat{w}_4 = (3 - \hat{p}_3^2)/[2\hat{p}_2^2(\hat{p}_2^2 - \hat{p}_3^2)] \\ \hat{w}_5 = (3 - \hat{p}_2^2)/[2\hat{p}_3^2(\hat{p}_3^2 - \hat{p}_2^2)]. \end{cases} \quad (37)$$

If $\hat{p}_3 = \hat{p}_2$, for Eq. (36) to remain valid, we must have $\hat{p}_3 = \hat{p}_2 = \sqrt{3}$. Then, $\hat{w}_3 = \frac{2}{3}$, $\hat{w}_4 = \hat{w}_5 = \frac{1}{6}$.

Note that the points $\hat{p}_1, \hat{p}_2$ and $\hat{p}_3$ are tunable parameters. When the locations of these points are given, the weights can be determined from Eqs. (35) to (37).

Remark 3.1. The moment matching method has been used to analyze (Julier & Uhlmann, 2004) or extend the UT (Tenne & Singh, 2003). The main difference of this paper's method from the methods used in those references is that only univariate moment matching is needed in this paper and then the resultant univariate point sets and weights are extended to the n -dimensional point set and weights using the sparse-grid method. Thus, the computations of the points and weights are greatly simplified.

Remark 3.2. Other univariate point-selection methods can be used if the condition of **Theorem 3.1** is satisfied. For example, we can use $4l - 1$ symmetric points for the univariate quadrature point set with the accuracy level l , which can match the $(4l - 2)$ th order moments. Increasing the number of univariate quadrature points increases the accuracy level of the univariate integral approximation. But it may not be sufficient to increase the accuracy level of the n -dimensional SGQ. It is worth noting that the univariate I_l with the accuracy level l requires the minimum number of l univariate quadrature points. In fact, one can choose l symmetric univariate GHQ points and weights as the univariate quadrature points and weights to satisfy the level- l requirement. Therefore, the univariate GHQ is a subset of the univariate quadrature proposed in this paper. The sparse Gauss–Hermite Quadrature (Jia, Xin, & Cheng, 2011) for multi-dimensional problems is a subset of the SGQ since they are based on the same sparse-grid method.

3.3. Multi-dimensional quadrature points and weights using the sparse-grid quadrature rule

To better illustrate the multi-dimensional quadrature points and weights, we show how to use the sparse-grid method, i.e. the Smolyak rule to construct $X_{2,3}$ ($n = 2, L = 3$) from the univariate quadrature point sets X_1, X_2 and X_3 containing 1, 3, and 5 symmetric points, respectively, as shown in Fig. 1.

From Eq. (28), q can be 1 or 2 and then we have $\mathbf{N}_1^2 = \{(1, 2), (2, 1)\}$ and $\mathbf{N}_2^2 = \{(1, 3), (2, 2), (3, 1)\}$. The first accuracy level combination in $\mathbf{N}_1^2$ is (1, 2), i.e. $i_1 = 1$ and $i_2 = 2$. This implies the tensor product $X_1 \otimes X_2$ and leads to 3 points along the horizontal axis.

Similarly, the second combination in $\mathbf{N}_1^2$ is (2, 1), i.e. $i_1 = 2$ and $i_2 = 1$. It generates the tensor product $X_2 \otimes X_1$ and 3 points along the vertical axis. Note that in the tensor products of $X_1 \otimes X_2$ and $X_2 \otimes X_1$, there are two new points added to the point set along two axes and one repeated point at the origin contributed from X_1 , which is regarded as an old point. Other tensor products in $\mathbf{N}_2^2$ can be generated in the same way. The final sparse-grid point set $X_{2,3}$ yields 17 points as shown on the right of Fig. 1.

The SGQ points and weights are generated in Algorithm 1.

Algorithm 1: Generate SGQ Points and Weights

```

 $[\chi, W] = \text{SGQ}[n, L, \hat{p}]$ 
( $\chi$ : SGQ point set with the element of  $\chi_i$ ;  $W$ : weight sequence with the element of  $W_i$ ;  $\hat{p}$  is the univariate SGQ point set)
For the points  $\hat{p}_i \in \hat{p}$ , determine the corresponding weights  $\hat{w}_s \in \hat{w}$ , ( $s \in \mathbb{N}$ ) for univariate quadrature rules  $I_l$  using Eq. (33) where  $\hat{w}$  is the univariate SGQ weight set.
FOR  $q = L - n : L - 1$ 
  Determine  $\mathbf{N}_q^n$ 
  FOR each element  $\mathcal{E} = (i_1, \dots, i_n)$  in  $\mathbf{N}_q^n$ , form
     $X_{i_1} \otimes X_{i_2} \dots \otimes X_{i_n}$ 
    FOR each point  $[\hat{p}_{s_1}, \dots, \hat{p}_{s_j}, \dots, \hat{p}_{s_n}]^T$  ( $\hat{p}_{s_j} \in \hat{p}$ ) in
       $X_{i_1} \otimes \dots \otimes X_{i_j} \dots \otimes X_{i_n}$ 
      IF the point is new, add it to  $\chi$ , assign a new index  $i$  to this point and calculate the weight of  $\chi_i$  as
        
$$W_i = (-1)^{L-1-q} C_{n-1}^{L-1-q} \prod_{j=1}^n \hat{w}_{s_j} \quad (\hat{w}_{s_j} \in \hat{w}) \quad (38)$$

      ELSE (the point already exists)
        update the old weight by
        
$$W_i = W_i + (-1)^{L-1-q} C_{n-1}^{L-1-q} \prod_{j=1}^n \hat{w}_{s_j} \quad (39)$$

      END IF
    END FOR
  END FOR
END FOR

```

3.4. Relationship among the SGQ rule, the UT, and the GHQ rule

The following theorem shows that the sigma-points and weights generated by the UT is a subset of the SGQ.

Theorem 3.2. *The points and weights generated by the UT are identical to the points and weights generated by the SGQ rule with level-2 accuracy if one point and three symmetric points are used for the level-1 and level-2 univariate quadrature point set respectively.*

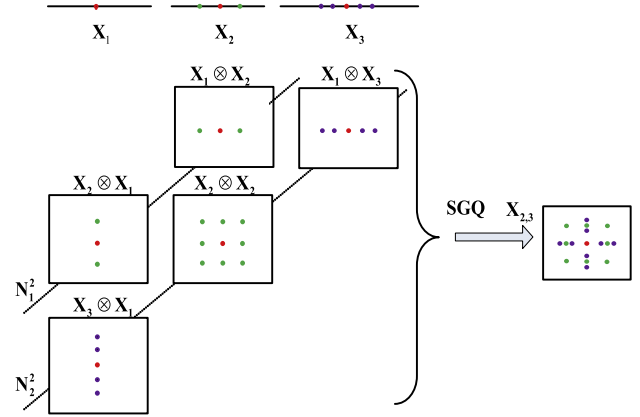


Fig. 1. Two-dimensional SGQ with accuracy level-3.

Proof. For the level-1 univariate quadrature rule I_1 , the point set is $\{0\}$ with the corresponding weight of 1. For the level-2 univariate quadrature I_2 , the univariate quadrature point set is $\{-\hat{p}_1, 0, \hat{p}_1\}$ with the corresponding weight sequence $(\hat{w}_2, \hat{w}_1, \hat{w}_2)$.

Since the accuracy level of the SGQ is $L = 2$, the value of q in $\mathbf{N}_q^n$ can be 0 or 1. When $q = 0$, $\mathbf{N}_0^n = \left\{ \underbrace{(1, 1, \dots, 1, 1)}_{n \text{ elements}} \right\}$, the n -dimensional SGQ point corresponding to $\mathbf{N}_0^n$ will be $[0, 0, \dots, 0, 0]^T$ with the corresponding weight of $(-1)^{2-1-0} \times C_{n-1}^{2-1-0} \times \underbrace{(1 \times 1 \times \dots \times 1 \times 1)}_{\text{product of weights}} = -(n-1)$ (using Eq. (38)).

When $q = 1$, the set of accuracy level sequences becomes as given in Box I.

Corresponding to the sequence (2, 1, ..., 1, 1), there are three n -dimensional SGQ points $[-\hat{p}_1, 0, \dots, 0, 0]^T$, $[0, 0, \dots, 0, 0]^T$, and $[\hat{p}_1, 0, \dots, 0, 0]^T$. The weights corresponding to $[-\hat{p}_1, 0, \dots, 0, 0]^T$ and $[\hat{p}_1, 0, \dots, 0, 0]^T$ are the same, i.e. $W_2 = (-1)^{2-1-1} \times C_{n-1}^{2-1-1} \times \underbrace{(1 \times 1 \times \dots \times 1 \times \hat{w}_2)}_{\text{product of weights}} = \hat{w}_2$ (using Eq. (38)).

Moreover, for this sequence, the SGQ can match the polynomials of the form $(ax_1^2 + bx_1^2 + cx_1 + d)x_2^{k_1}x_3^{k_2} \dots x_n^{k_n}$, where a, b, c, d are real numbers and $k_1, \dots, k_n$ can be 0 or 1, because I_2 is exact for all univariate polynomials of the order up to $2 \times 2 - 1 = 3$ and I_1 is exact for all univariate polynomials of the order up to $2 \times 1 - 1 = 1$ according to Theorem 3.1. Similarly, for other sequences in $\mathbf{N}_1^n$ like (1, ..., 1, 2, 1, ..., 1), there are three n -dimensional SGQ points $[0, \dots, 0, -\hat{p}_1, 0, \dots, 0]^T$, $[0, \dots, 0, 0, 0, \dots, 0]^T$, and $[0, \dots, 0, \hat{p}_1, 0, \dots, 0]^T$. The weights corresponding to $[0, \dots, 0, -\hat{p}_1, 0, \dots, 0]^T$ and $[0, \dots, 0, \hat{p}_1, 0, \dots, 0]^T$ are the same, i.e. $W_2 = \hat{w}_2$ (using Eq. (38)). There are n such combinations of accuracy level sequences in $\mathbf{N}_1^n$ with the similar calculation of n -dimensional SGQ points and weights. Note that since the point $[0, \dots, 0, 0, 0, \dots, 0]^T$ is a repeated point appearing once in $\mathbf{N}_0^n$ and n times in $\mathbf{N}_1^n$, Eq. (39) is used to calculate its weight (see Box II).

To summarize it, the points and weights of the SGQ with level-2 accuracy are

$$\gamma_i = \begin{cases} [0, 0, \dots, 0, 0]^T; & i = 1 \\ \hat{p}_1 \mathbf{e}_{i-1}; & 2 \leq i \leq n+1 \\ -\hat{p}_1 \mathbf{e}_{i-n-1}; & n+2 \leq i \leq 2n+1 \end{cases} \quad (40)$$

$$\mathbf{N}_1^n = \left\{ \underbrace{(2, 1, \dots, 1, 1), (1, 2, \dots, 1, 1), \dots, (1, \dots, 1, 2, 1, \dots, 1), \dots, (1, 1, \dots, 1, 2)}_{n \text{ elements}} \right\}.$$

Box I.

$$\begin{aligned} W_1 &= (-1)^{2-1-0} \times C_{n-1}^{2-1-0} \times \underbrace{\left(\overbrace{1 \times 1 \times \dots \times 1 \times 1}^{n \text{ elements}} \right)}_{\text{product of weights}} + (-1)^{2-1-1} \times C_{n-1}^{2-1-1} \\ &\times \left\{ \underbrace{\left(\overbrace{\hat{w}_1 \times 1 \times 1 \times \dots \times 1}^{n-1 \text{ elements}} \right)}_{\text{product of weights}} + \underbrace{\left(\overbrace{1 \times \hat{w}_1 \times 1 \times 1 \times \dots \times 1}^{n-2 \text{ elements}} \right)}_{\text{product of weights}} + \dots + \underbrace{\left(\overbrace{1 \times 1 \times \dots \times 1 \times \hat{w}_1}^{n-1 \text{ elements}} \right)}_{\text{product of weights}} \right\} \\ &= -(n-1) + n \cdot \hat{w}_1. \end{aligned}$$

Box II.

and

$$W_i = \begin{cases} -(n-1) + n \cdot \hat{w}_1 & i = 1 \\ \hat{w}_2 & i = 2 \dots 2n+1 \end{cases} \quad (41)$$

respectively. If we choose $\hat{p}_1 = \sqrt{n+\kappa}$, by Eq. (35), $\hat{w}_1 = 1 - \frac{1}{(n+\kappa)}$ and $\hat{w}_2 = \frac{1}{2(n+\kappa)}$. Hence

$$W_i = \begin{cases} -(n-1) + n \cdot \left(1 - \frac{1}{(n+\kappa)} \right) = \frac{\kappa}{n+\kappa} & i = 1 \\ \frac{1}{2(n+\kappa)} & i = 2 \dots 2n+1. \end{cases} \quad (42)$$

Comparing Eqs. (40) and (42) with the sigma points and weights of the UT in Eq. (24), they are identical. $\square$

Theorem 3.2 implies that the UKF is a subset of the SGQF at the accuracy level-2. The SGQF can achieve higher accuracy levels than the UKF by increasing L .

From Eq. (35), one can also see the relationship between the SGQ and the GHQ. If one chooses $\hat{p}_1 = \sqrt{3}$, the univariate quadrature point set is exactly the same as the univariate GHQ point set when 3 points ($m_q = 3$) are used for the GHQ. It can be verified by the algorithm introduced in Section 2.2.1. However, the methods to extend the univariate quadrature to the multi-dimensional SGQ and multidimensional GHQ are different. The multi-dimensional GHQ uses the direct tensor product rule whereas the SGQ uses the sparse-grid method. The direct tensor product rule generates much more points than the sparse-grid method. When the SGQ point set and the multi-dimensional GHQ point set are generated from the same univariate GHQ points, the former is a subset of the latter. However, the weights on the points shared by the SGQ and multi-dimensional GHQ are different. Recall the accuracy of SGQ is given by Theorem 3.1 and the accuracy of GHQ is given in Section 2.2.1.

3.5. The number, locations, and weights of the SGQ points

The upper bound of the number of sparse-grid points using the Gauss quadrature has been analyzed in Heiss and Winschel (2008) when the univariate quadrature rule I_l has l points. In the following proposition, we analyze the number of SGQ points in an explicit polynomial form.

Proposition 3.1. Given the accuracy level L ($1 < L \leq n+1$), the number of SGQ points increases polynomially with the dimension n and the highest order of this polynomial is $L-1$.

Proof. When $L \leq n$, q can be $0, 1, \dots, L-1$. Thus, $\mathbf{N}_q^n$ can be $\mathbf{N}_0^n, \dots, \mathbf{N}_{L-1}^n$. When $L = n+1$, q can be $1, \dots, L-1$. For $\mathbf{N}_q^n$, each element sequence in it contains at most q elements that are greater than 1. If it contains $q+1$ elements greater than 1, then we have $\sum_{j=1}^n i_j \geq \underbrace{n-q-1}_{\text{element}=1} + 2 \times \underbrace{(q+1)}_{\text{element}=2} > n+q$, which contradicts the definition of $\mathbf{N}_q^n$ in (27). The largest q for $\mathbf{N}_q^n$ to be a nonempty set is $L-1$.

When $q = L-1$, each accuracy level sequence in $\mathbf{N}_{L-1}^n$ contains at most $L-1$ elements greater than 1. If there are $L-1$ elements greater than 1, there are C_n^{L-1} kinds of combinations of such accuracy level sequences. If there are p ($1 \leq p < L-1$) elements greater than 1, there are C_n^p kinds of combinations of such accuracy level sequences.

When $0 \leq q < L-1$, each accuracy level sequence in $\mathbf{N}_q^n$ contains at most q elements greater than 1. If there are q elements greater than 1, there are C_n^q kinds of combinations of such accuracy level sequences. If there are p ($1 \leq p < q$) elements greater than 1, there are C_n^p kinds of combinations of such accuracy level sequences.

For each accuracy level sequence, the number of new points is bounded. The total number of points is a linear combination of the products of the number of new points generated from each accuracy level sequence and the associated combinatorial C_n^s ($s = p$ or q), which is a polynomial in terms of the dimension n . Hence, when the accuracy level L is given, the total number of points is a polynomial in terms of n and the highest order of this polynomial is $L-1$, which is determined by C_n^{L-1} . $\square$

Remark 3.3. In general, the accuracy level L is much smaller than the dimension n . Higher accuracy level SGQ needs more points and increases the computation load unnecessarily when a low accuracy level can provide acceptable performance. In addition, L should be greater than 1 since the SGQ point set for $L=1$ contains only one point, which is not sufficient for approximation.

The number of SGQ points can be calculated by analyzing the number of points corresponding to all different accuracy level sequences in $\mathbf{N}_q^n$. However, we can use the highest accuracy level

set to calculate the number of points when the condition in the following proposition is satisfied.

Proposition 3.2. *If $X_1 \subset X_2$, $1 < q < n$, the SGQ points corresponding to the accuracy level set $\mathbf{N}_{q-1}^n$ are contained in the SGQ point set corresponding to $\mathbf{N}_q^n$.*

Proof. By the definition, $\mathbf{N}_{q-1}^n = \{\mathcal{E} : \sum_{j=1}^n i_j = n + q - 1\}$ and $\mathbf{N}_q^n = \{\mathcal{E} : \sum_{j=1}^n i_j = n + q - 1 + 1\}$. If the sequence $(i_1, i_2, \dots, i_j, \dots, i_n)$ belongs to $\mathbf{N}_{q-1}^n$, then the sequence $(i_1, i_2, \dots, i_j + 1, \dots, i_n)$ belongs to $\mathbf{N}_q^n$. For $\mathbf{N}_q^n$, each element sequence in it contains at most q elements that are greater than 1, which has been discussed in the proof of Proposition 3.1. Because $1 < q < n$ and there are at most q elements greater than 1 in the accuracy level sequence, we can always find at least one $i_j = 1$, $1 \leq j \leq n$ such that the points corresponding to $(i_1, i_2, \dots, i_j, \dots, i_n)$ are generated by the tensor product rule $X_{i_1} \otimes X_{i_2} \otimes \dots \otimes X_1 \otimes \dots \otimes X_{i_n}$ whereas the points corresponding to $(i_1, i_2, \dots, i_j + 1, \dots, i_n)$ are generated by the tensor product rule $X_{i_1} \otimes X_{i_2} \otimes \dots \otimes X_2 \otimes \dots \otimes X_{i_n}$. Because $X_1 \subset X_2$, any point generated by $\mathbf{N}_{q-1}^n$ will be contained in $\mathbf{N}_q^n$. $\square$

Note that the condition $1 < q < n$ in Proposition 3.2 is generally true for a normal problem since the largest value of q is $L - 1$ and L is usually much smaller than n as mentioned in the Remark 3.3. So one can usually calculate the SGQ points determined by $\mathbf{N}_{L-1}^n$.

The points and weights with level-2 accuracy are shown in Eqs. (40) and (41). For convenience, we derive the expressions for the n -dimensional SGQ points and weights with level-3 accuracy using the same procedure as the proof of Theorem 3.2. Note that these points and weights can be automatically obtained using Algorithm 1 as well. We use 3 and 5 (or 3) distinct symmetric points for the univariate quadrature point set with accuracy levels 2 and 3, respectively. The n -dimensional SGQ points and weights with accuracy level-3 may have three different cases depending on the locations of the univariate quadrature points. The equations of computing the points and weights for these three different cases are given in the Appendix.

3.6. Discussions

There are several important advantages of the SGQ over other point-based methods.

(1) SGQ with different accuracy levels can be implemented using the same sparse-grid framework. When univariate quadrature points and weights of different accuracy levels are given, the multidimensional SGQ point set and weights for a given accuracy level can be uniquely determined. Theoretically, if the nonlinear function $\mathbf{f}$ can be approximated sufficiently well by polynomials, then the integral $\int_{-\infty}^{\infty} \mathbf{f}(\mathbf{x}) N(\mathbf{x}; \mathbf{0}, \mathbf{I}) d\mathbf{x}$ can be calculated with sufficient accuracy by increasing the level of SGQ. Other point-based methods are difficult to extend to achieve higher level accuracy. For example, the integral $\int_{-\infty}^{\infty} (1 + \sum x_i^2)^n N(\mathbf{x}; \mathbf{0}, \mathbf{I}) d\mathbf{x}$ (x_i is the i th element of the vector $\mathbf{x}$) can be calculated exactly by the SGQ with accuracy level-4 and level-5 when n equals 3 or 4 whereas UKF or CKF is difficult to achieve this accuracy.

(2) The number of SGQ points for a fixed accuracy level increases polynomially with the dimension. The classical method to extend the one-dimensional integral approximation rule to the multidimensional problem is the direct tensor product rule. The multidimensional GHQ rule uses this method and the number of the multi-dimensional GHQ points increases exponentially with the dimension, which severely limits the applicability of this rule. Using the SGQ points based on the sparse-grid method, however,

the SGQF algorithm needs far fewer points. Thus, it alleviates the computational load and is very efficient for high dimensional estimation problems.

(3) The tunable parameters $\hat{p}_1, \hat{p}_2, \hat{p}_3, \dots$ make the SGQF more flexible than the GHQF or the sparse GHQF. Since the point-based methods evaluate the value of the nonlinear function at those points, it is understandable that the location of those points may affect the accuracy of the integral result. For different nonlinear functions, different point-sets with the same accuracy level may have very different performance. These tunable parameters can capture non-Gaussian pdfs to some extent. It is similar to the UKF that has an adjustable parameter κ to tune the performance depending on applications. In comparison with the GHQ rule, the locations of the GHQ points are fixed, which is not flexible for non-Gaussian problems. Instead of using fixed points, the SGQ can adjust the locations of the points. When the univariate GHQ rule is used as the univariate quadrature rule for the SGQ, SGQ will have the same points and weights as the Sparse Gauss Hermite Quadrature (SGHQ) rule (Jia et al., 2011). Compared with the SGHQ, the SGQ has tunable parameters. When selecting univariate quadrature points, the univariate GHQ points can be used for SGQ as the reference values of the tunable parameters. Moreover, it is worth noting that the SGQ can use nested sequences of quadrature rules, which means that the univariate quadrature point set with the lower accuracy level is a subset of the univariate quadrature point set with the higher accuracy level, i.e. $X_i \subset X_j$ if $i < j$. For example, in Case 2 of the level-3 SGQ (Appendix), the univariate quadrature point set $\{-\hat{p}_1, 0, \hat{p}_1\}$ at level-2 is nested in the univariate quadrature point set $\{-\hat{p}_3, -\hat{p}_2, 0, \hat{p}_2, \hat{p}_3\}$ at level-3 because of $\hat{p}_1 = \hat{p}_2$. The nested structure can reduce the total number of the SGQ points when extended to the multi-dimensional point set using the sparse-grid method compared with other non-nested SGQs like SGHQ.

4. Numerical results and analysis

In this section, the performance of the SGQF is demonstrated by the orbit estimation problem and is compared with EKF, UKF, CKF, and GHQF.

The orbit estimation problem is to obtain the accurate satellite position and velocity from noisy observations. The dynamic and measurement models are given below.

The low-earth orbit (LEO) satellite dynamics can be described by Crassidis and Junkins (2004)

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3} \mathbf{r} + \mathbf{a}_G + \mathbf{a}_D + \mathbf{v} \quad (43)$$

where $\mathbf{r} = [x, y, z]^T$ is the position of the satellite in the inertial coordinate frame ($\mathbf{I} - \mathbf{J} - \mathbf{K}$); $r = \sqrt{x^2 + y^2 + z^2}$; $\mathbf{v}$ is the white Gaussian process noise; $\mathbf{a}_G$ is the instantaneous acceleration due to the J_2 perturbation (Prussing & Conway, 1993). J_2 is the dimensionless second zonal harmonic that quantifies the major oblateness effect of the Earth. J_2 perturbation may cause a noticeable precession of the LEO satellite orbits. $\mathbf{a}_D$ is the atmospheric drag with

$$\mathbf{a}_D = -\frac{1}{2} \frac{C_d A}{m_s} \rho_d v_{rel} \mathbf{v}_{rel} \quad (44)$$

where ρ_d is the atmospheric density; m_s is the satellite's mass; C_d is the drag coefficient; $\mathbf{v}_{rel}$ is the velocity vector relative to the rotating atmosphere and A is cross-sectional area (Vallado, 2001). In this paper, we only consider J_2 and atmospheric drag perturbations. The lunar/solar gravity perturbation is very small and is negligible for the low-earth orbit (LEO) satellite. All the unmodeled perturbations are regarded as the process noise.

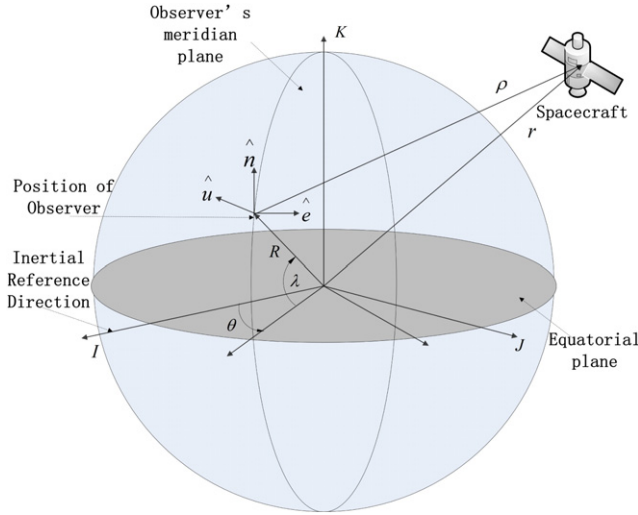


Fig. 2. Illustration of the observation geometry.

The measurement model is described by Crassidis and Junkins (2004)

$$\begin{cases} az = \tan^{-1} \left(\frac{\rho_e}{\rho_n} \right) + n_{az} \\ el = \tan^{-1} \left(\frac{\rho_u}{\sqrt{\rho_e^2 + \rho_n^2}} \right) + n_{el} \\ \|\rho\| = \sqrt{\rho_u^2 + \rho_e^2 + \rho_n^2} + n_\rho \end{cases} \quad (45)$$

where the azimuth (az), the elevation (el), and the range $\rho = [\rho_u \ \rho_e \ \rho_n]^T$ can be measured by the radar site on the ground with respect to the local observer coordinate system, ($\hat{u} - \hat{e} - \hat{n}$; “up, east and north”). The range can be related to the position vector in the inertial frame ($I - J - K$) by the coordinate transformation given by Eq. (46).

$$\begin{bmatrix} \rho_u \\ \rho_e \\ \rho_n \end{bmatrix} = \begin{bmatrix} \cos \lambda & 0 & \sin \lambda \\ 0 & 1 & 0 \\ -\sin \lambda & 0 & \cos \lambda \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x - \|\mathbf{R}\| \cos \lambda \cos \theta \\ y - \|\mathbf{R}\| \cos \lambda \sin \theta \\ z - \|\mathbf{R}\| \sin \lambda \end{bmatrix} \quad (46)$$

where $\|\mathbf{R}\| = 6378.1363$ km is the earth radius; λ and θ are the latitude and local sidereal time of the observer respectively; n_{az}, n_{el}, n_ρ are the white Gaussian measurement noise. The geometry of the observation model is shown in Fig. 2.

Generally, it is impossible to track a LEO satellite all the time from a single radar station. In the assumed simulation scenario, the reasonable track time is about 5 min. Hence, the simulation time is set to be 5 min and the measurement period is 5 s. The step size of 0.1 s and the fourth-order Runge–Kutta algorithm are used for dynamic propagation (Lee & Alfriend, 2007). The latitude and the longitude of the radar site are -10.749° and -70.5983° , respectively. The initial simulation time is March 29, 2011, 17:00:00. Moreover, the atmospheric drag parameters are $C_d = 2.2$, $A/m_s = 0.02$ m²/kg, and the atmospheric density ρ_d is calculated by the exponential model (Vallado, 2001).

The state variables of this orbit estimation problem is

$$\mathbf{x} = \begin{bmatrix} (\mathbf{x}^p)^T & (\mathbf{x}^v)^T \end{bmatrix}^T = \begin{bmatrix} x & y & z & \dot{x} & \dot{y} & \dot{z} \end{bmatrix}^T.$$

The initial true state value is assumed to be

$$\mathbf{x}_0 = \begin{bmatrix} (\mathbf{x}_0^p)^T & (\mathbf{x}_0^v)^T \end{bmatrix}^T$$

Table 1
Parameters of SGQFs.

	$\hat{p}_1$	$\hat{p}_2$	$\hat{p}_3$
1st SGQF (Case 1)	$\sqrt{3}$	$\sqrt{3}$	$\sqrt{3}$
2nd SGQF (Case 2)	1.71	1.71	2.50
3rd SGQF (Case 2)	1	1	2.5
4th SGQF (Case 2)	0.5	0.5	2
5th SGQF (Case 3)	1.71	1.00	2.50
6th SGQF (Case 3)	1	2.5	1.5
7th SGQF (Case 3)	0.5	2	1

where $(\mathbf{x}_0^p)^T = [6949.599783, 1045.733299, 64.918535]$ km

and $(\mathbf{x}_0^v)^T = [-0.902571, 5.697655, 4.841182]$ km/s

are the initial true position and velocity of the satellite respectively. The above orbital parameters including the initial conditions, the radar site, and the track time are verified to be feasible using the STK[®], a well-accepted software tool for high-fidelity modeling of satellite orbits.

The level-2 and level-3 SGQFs are applied to this orbit estimation problem and compared with the EKF, the UKF, the CKF, and the GHQF. The level-2 SGQ points and weights are computed using Eqs. (40) and (41) and the level-3 SGQ points and weights are computed using Eqs. (47)–(52).

There are three cases of level-3 SGQFs as discussed in the Appendix. Case 1 does not have any tunable parameters and uses the GHQ points $\hat{p}_1 = \hat{p}_2 = \hat{p}_3 = \sqrt{3}$; Case 2 has two tunable parameters, $\hat{p}_1 (= \hat{p}_2)$ and $\hat{p}_3$; and Case 3 has three tunable parameters, $\hat{p}_1, \hat{p}_2$, and $\hat{p}_3$. The SGQF parameter $\hat{p}_1$ is used in both level-2 and level-3 SGQFs. The other two parameters are used only in the level-3 SGQFs. In other words, the level-2 SGQF only has one tunable parameter $\hat{p}_1$ while the level-3 SGQFs have up to three tunable parameters.

To investigate the effect of the SGQ parameters on the filter performance, seven SGQFs with different parameters are tested. The GHQ points $\hat{p}_1 = \hat{p}_2 = \hat{p}_3 = \sqrt{3}$ are used as the reference points about which $\hat{p}_1, \hat{p}_2$, and $\hat{p}_3$ are varied and given in Table 1. Note that, for level-2 SGQFs, the SGQ points and weights are only determined by $\hat{p}_1$, and for level-3 SGQFs the locations and weights of most points are determined by $\hat{p}_1$ from the Eqs. (47)–(52). Therefore, the influence of $\hat{p}_1$ is more significant than the other two points.

In the following, the root-mean square errors (RMSEs) of SGQFs over 50 runs are compared with those of the EKF, UKF, CKF, and GHQF. The RMSEs of the position and velocity estimates at time k is defined as

$$\begin{aligned} \text{RMSE}_{\text{pos}}(k) &= \sqrt{\frac{1}{N_{mc}} \sum_{n=1}^{N_{mc}} ((x_k - \hat{x}_k)^2 + (y_k - \hat{y}_k)^2 + (z_k - \hat{z}_k)^2)} \\ \text{RMSE}_{\text{vel}}(k) &= \sqrt{\frac{1}{N_{mc}} \sum_{n=1}^{N_{mc}} ((\dot{x}_k - \dot{\hat{x}}_k)^2 + (\dot{y}_k - \dot{\hat{y}}_k)^2 + (\dot{z}_k - \dot{\hat{z}}_k)^2)} \end{aligned}$$

where $N_{mc} = 50$ is the number of simulation runs; $[\hat{x}_k \ \hat{y}_k \ \hat{z}_k]^T$ and $[\dot{\hat{x}}_k \ \dot{\hat{y}}_k \ \dot{\hat{z}}_k]^T$ are the estimated position and velocity at time k , respectively.

Two different scenarios, small initial estimation errors and large initial estimation errors, are used to test the filter performance.

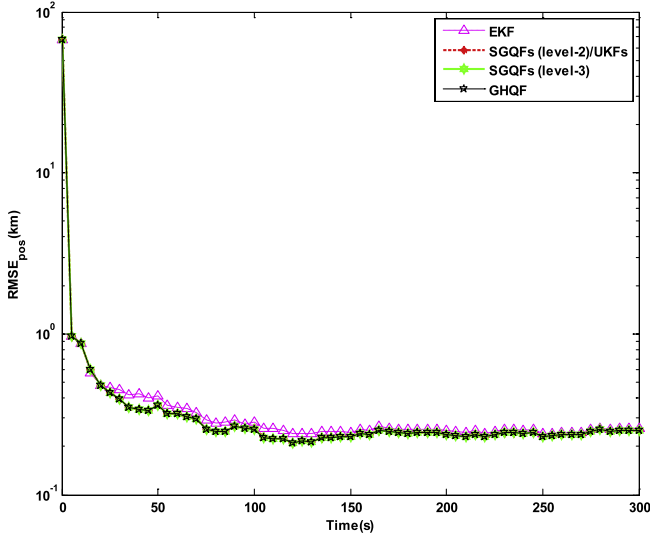


Fig. 3. RMSEs of the position with small initial estimation errors.

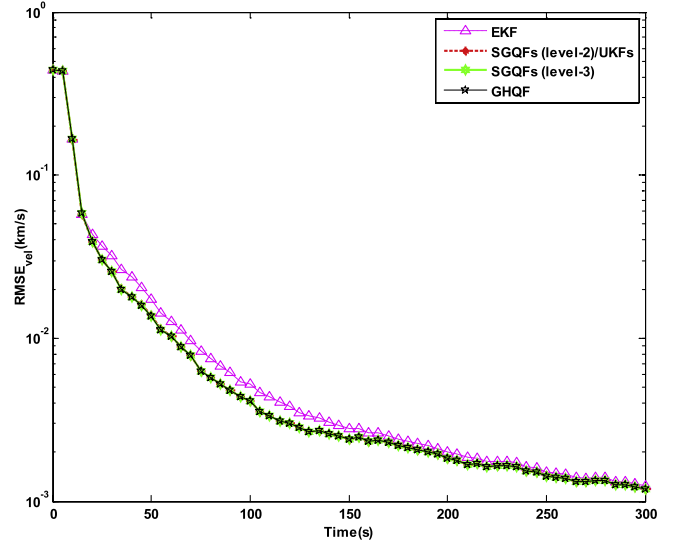


Fig. 4. RMSEs of the velocity with small initial estimation errors.

4.1. The first scenario: small initial estimation errors

In this scenario, the initial estimates are assumed with small errors

$$(\hat{\mathbf{x}}_0^p)^T = [6989.157085, 1061.379082, 117.351971] \text{ km}$$

$$\text{and } (\hat{\mathbf{x}}_0^v)^T = [-0.899065, 5.291087, 5.005861] \text{ km/s.}$$

The initial covariance is

$$\mathbf{P}_0 = \text{diag} \left(\begin{bmatrix} 10^2, 10^2, 10^2 \end{bmatrix} (\text{km})^2, \right. \\ \left. \begin{bmatrix} 10^{-2}, 10^{-2}, 10^{-2} \end{bmatrix} (\text{km/s})^2 \right).$$

The reasonable process and measurement noise covariances are assumed to be (Lee & Alfriend, 2007)

$$\mathbf{Q}_k = \text{diag} ([0, 0, 0, 10^{-17} (\text{km/s})^2, 10^{-17} (\text{km/s})^2, 10^{-17} (\text{km/s})^2])$$

and $\mathbf{R}_k = \text{diag} ([(0.015^\circ)^2, (0.015^\circ)^2, 0.025^2 (\text{km})^2])$ respectively.

The SGQFs (level-2 and level-3) using the parameters in Table-1 are tested and compared with the EKF, UKF/CKF, and GHQF. Since κ is a tunable parameter for UKF, a range of $\kappa = -3, -2, -1, 0, 1, 2, 3$ are used for the test. The position and velocity RMSEs are shown in Figs. 3 and 4 respectively. Because there is no discernible difference among all SGQFs (level-2 or level-3) and UKFs are identical to level-2 SGQFs with $\hat{p}_1 = \sqrt{n + \kappa}$ (Theorem 3.2), only one of the level-2 SGQFs and one of the level-3 SGQFs are given in Figs. 3 and 4. The simulation results show that SGQFs are slightly more accurate than EKF. The level-2 SGQFs, level-3 SGQFs, and GHQF all have so close performance that they are indistinguishable. Therefore, the point-based methods (UKFs, SGQFs, and GHQF) do not show significant advantages over EKF when the initial estimation errors are small.

4.2. The second scenario: large initial estimation errors

In this scenario, the initial estimates are assumed with large errors

$$(\hat{\mathbf{x}}_0^p)^T = [7252.009273, 1358.407862, 383.904071] \text{ km}$$

$$\text{and } (\hat{\mathbf{x}}_0^v)^T = [-0.613101, 5.991868, 5.138553] \text{ km/s.}$$

The initial covariance is

$$\mathbf{P}_0 = \text{diag} \left(\begin{bmatrix} 10^4, 10^4, 10^4 \end{bmatrix} (\text{km})^2, \right. \\ \left. \begin{bmatrix} 10^{-2}, 10^{-2}, 10^{-2} \end{bmatrix} (\text{km/s})^2 \right).$$

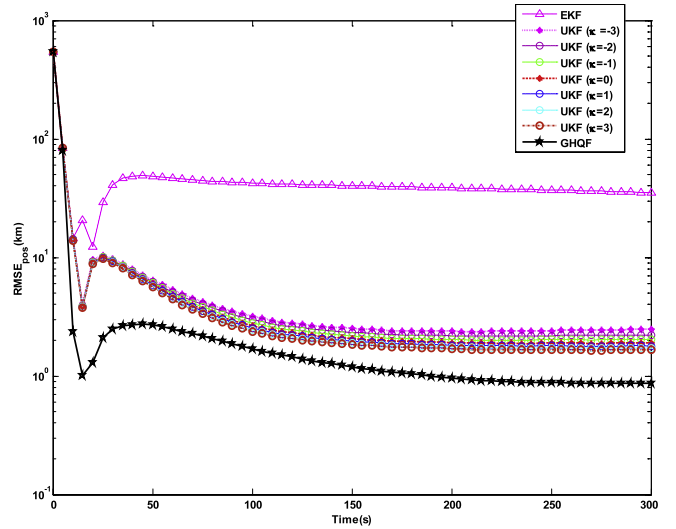


Fig. 5. RMSEs of the position with large initial estimation errors: EKF, UKFs, and GHQF.

4.2.1. Results of UKFs and level-2 SGQFs

In this section, we compare the SGQF at accuracy level-2 with EKF and GHQF. The first group of tested level-2 SGQFs is the UKFs with a range of the tunable parameters $\kappa = -3, -2, -1, 0, 1, 2, 3$. Recall that the UKF is identical to the level-2 SGQF with $\hat{p}_1 = \sqrt{n + \kappa}$ (Theorem 3.2 and Eq. (40)). The role of the tunable parameter κ for UKF is the same as the tunable parameter $\hat{p}_1$ for SGQF.

The RMSEs of the position and velocity estimates for EKF, UKFs, and GHQF are shown in Figs. 5 and 6, respectively.

As can be seen, UKFs and GHQF are much more accurate than EKF for large initial estimation errors. Additionally, all the UKFs for different κ parameters exhibit close performance. Note that the suggested parameter of κ is $\kappa = 3 - n = -3$ (Julier et al., 2000), and the CKF is identical to the UKF when $\kappa = 0$. For this particular application, UKF has the best performance when $\kappa = 3$. The CKF is slightly better than the UKF with the suggested value of $\kappa = -3$. GHQF is obviously more accurate than all UKFs.

In the above group of tested UKFs or level-2 SGQFs, the parameter $\hat{p}_1$ corresponding to κ used for UKF are all greater than or equal to $\sqrt{3}$. So in the second group of tested level-2 SGQFs,

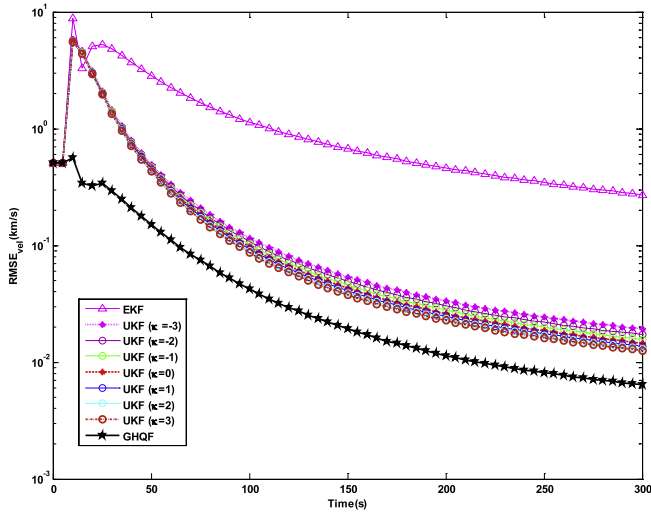


Fig. 6. RMSEs of the velocity with large initial estimation errors: EKF, UKFs, and GHQF.

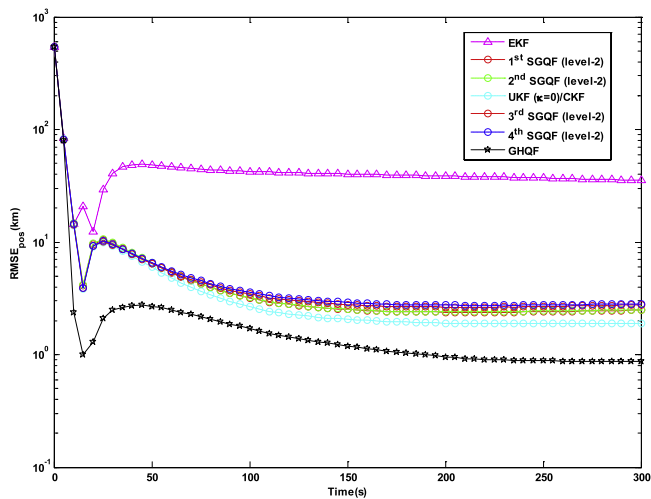


Fig. 7. RMSEs of the position with large initial estimation errors: EKF, level-2 SGQFs, and GHQF.

we use the values of $\hat{p}_1$ that are smaller than $\sqrt{3}$ for further comparison. These parameters of $\hat{p}_1$ correspond to the 1st SGQF to the 4th SGQF in Table 1. Since all the UKFs and CKF do not show much difference, we will select the CKF and the UKF with the suggested parameter $\kappa = -3$ as the references for comparison with other level-2 SGQFs, i.e. the 1st SGQF (level-2) to the 4th SGQF (level-2) in Table 1.

The RMSEs of the position and velocity for EKF, level-2 SGQFs, and GHQF are shown in Figs. 7 and 8. As can be seen, all level-2 SGQFs are much more accurate than the EKF but not as accurate as GHQF. Note that the 1st SGQF (level-2) is the same as the UKF with the suggested parameter $\kappa = -3$.

In summary, all the level-2 SGQFs including UKF and CKF with different parameter values demonstrate close performance, which is more accurate than EKF but not as accurate as GHQF.

4.2.2. Results of level-3 SGQFs

In this section, the level-3 SGQFs with different parameters of $\hat{p}_1$, $\hat{p}_2$, and $\hat{p}_3$ corresponding to the three cases in Table 1, is compared with EKF, UKF ($\kappa = 3$ with the best performance), and GHQF. The RMSEs of the position and velocity are shown in Figs. 9 and 10.

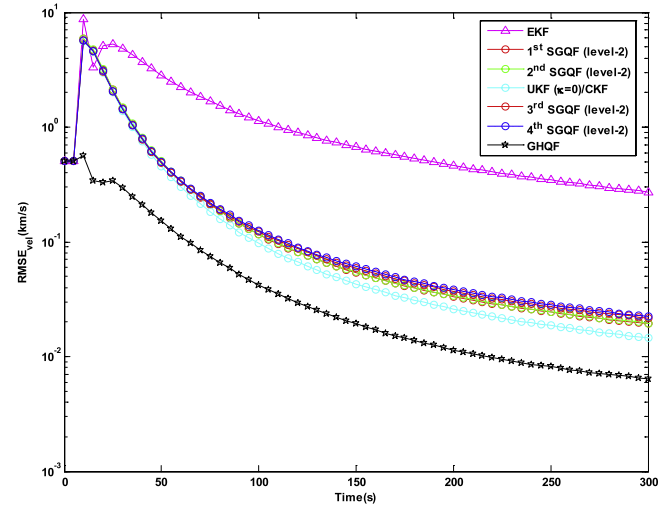


Fig. 8. RMSEs of the velocity with large initial estimation errors: EKF, level-2 SGQFs, and GHQF.

Table 2

Number of points for different SGQFs and GHQF.

	SGQF (Case 1)	SGQF (Case 2)	SGQF (Case 3)	GHQF $m_q = 3$
Level-2	13	13	13	729
Level-3	73	85	97	

It can be seen that all level-3 SGQFs are more accurate than the EKF and level-2 SGQF/UKF. All level-3 SGQFs are overlapped and have no discernible difference, which implies that these level-3 SGQFs are not sensitive to the SGQ parameters. Note that the level-2 SGQ can approximate the integral calculations up to the 3rd order polynomials whereas level-3 SGQ can be exact up to the 5th order polynomials (Theorem 3.1). From the simulation tests, it indicates that the 3rd order polynomial approximation is not accurate enough, which is the reason that one can see the performance difference (Section 4.2.1) using different parameters of $\hat{p}_1$ (or κ for UKF). However, the level-3 SGQ that is exact up to the 5th order polynomials is accurate enough to capture the nonlinear characteristics and initial estimate uncertainty of this orbit estimation problem. Note that no matter how the parameters $\hat{p}_1$, $\hat{p}_2$, $\hat{p}_3$ are tuned, the 5th order accuracy can be guaranteed and thus the performance will not exhibit noticeable difference.

Figs. 9 and 10 also show that the level-3 SGQFs have the same performance as the GHQF ($m_q = 3$). However, the level-3 SGQFs uses 73, 85, and 97 points for the Cases 1, 2, and 3, respectively whereas the GHQF uses $3^6 (= 729)$ points. We did not compare with the GHQF with higher accuracy because it would require a huge number of GHQ points, for example, $5^6 (= 15,625)$ for $m_q = 5$. The number of points for different SGQFs and GHQF are shown in Table 2, which indicates that the SGQF is computationally much more efficient than the GHQF.

From the above simulation results, Figs. 5–10, it can be seen that the level-3 SGQFs demonstrate the best performance and are good enough for this orbit estimation problem. It is worth noting that for other applications, higher level SGQF may be necessary.

In this paper, GHQ points are used as the reference points to tune the parameters of SGQ. If three point univariate GHQ is used for I_2 and I_3 , the reference points are $\hat{p}_1 = \hat{p}_2 = \hat{p}_3 = \sqrt{3}$. Higher accuracy GHQ points can be used for I_2 and I_3 as well. For example, 3 and 5 univariate GHQ points can be used for I_2 and I_3 , respectively, with the reference points being $\hat{p}_1 = \sqrt{3}$, $\hat{p}_2 = 1.3556$ and $\hat{p}_3 = 2.8570$. The parameters of SGQ can be tuned around these reference points. For SGQ of higher-than-three

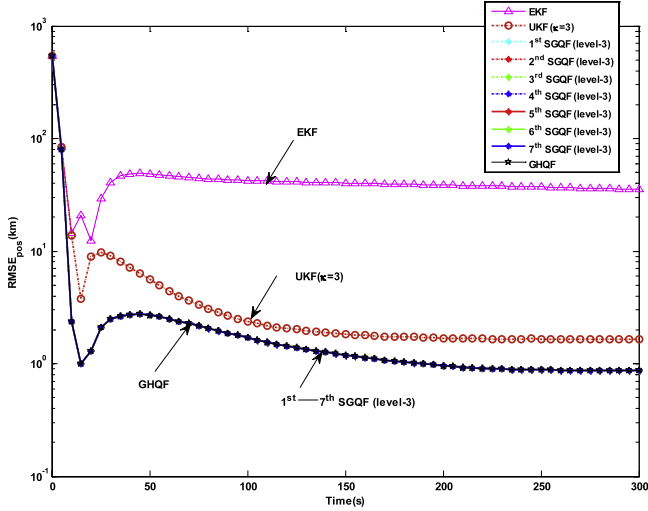


Fig. 9. RMSEs of the position with large initial estimation errors: EKF, UKF, level-3 SGQFs, and GHQF.

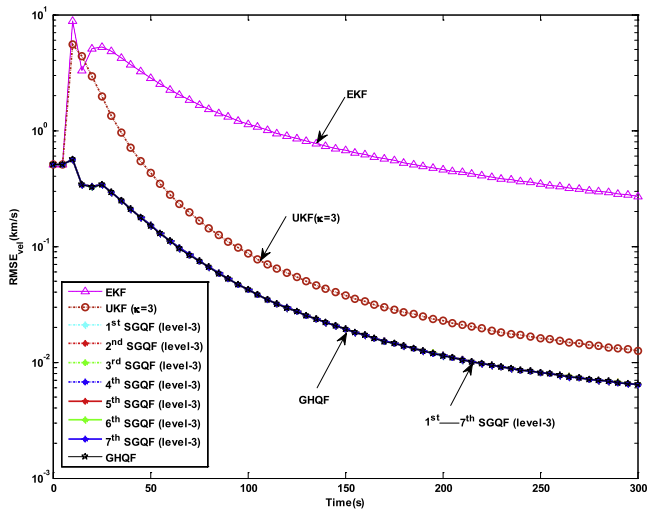


Fig. 10. RMSEs of the velocity with large initial estimation errors: EKF, UKF, level-3 SGQFs, and GHQF.

accuracy levels, which need more points to match higher moments, more than 3 univariate GHQ points should be used as the reference points. In addition, the GHQ points are optimal for the integrals when the integrand function follows the special form which is the multiplication of a polynomial function and the Gaussian pdf. For a general form which is the multiplication of a general nonlinear function and a non-Gaussian pdf, the GHQ points may not be optimal and the tunable parameter $\hat{p}_i$ can play an important role in the performance of the SGQF.

5. Conclusion

In this paper, a novel nonlinear filter, the Sparse-grid Quadrature Filter (SGQF), was developed using the moment matching method for the univariate integral approximation and the sparse-grid method for the multi-dimensional extension. Important relationships were established among the SGQF, UKF, and GHQF. It was proven that the UKF is a subset of the SGQF and that the UKF is equivalent to the SGQF with level-2 accuracy. In addition, the novel filter is more flexible to use than the UKF since higher accuracy levels can be easily achieved by moderately increasing the number of SGQ points. It was shown that the number of SGQ points

increases polynomially with the dimension. The SGQF is computationally much more efficient than the GHQF for multi-dimensional problems since it requires considerably fewer points. Because the univariate GHQ rule is a special case of the univariate SGQ rule, the sparse GHQF (Jia et al., 2011) is a special case of the SGQF. The simulation results in the orbit estimation application demonstrated that the novel filtering algorithm exhibits much better performance than the EKF, and when using the level-3 SGQ, is more accurate than the UKF and the CKF. Furthermore, with significantly less number of points, it exhibits the same performance as the GHQF.

Appendix

Equations to calculate the points and weights for level-3 SGQ:

Case 1: $\hat{p}_1 = \hat{p}_2 = \hat{p}_3$

In this case, the point set $\{-\hat{p}_1, 0, \hat{p}_1\}$ with the weights $(\hat{w}_2, \hat{w}_1, \hat{w}_2)$ is used for both level-2 and level-3 SGQ, and there are three variables, $\hat{p}_1$, $\hat{w}_1$ and $\hat{w}_2$. By Eqs. (34) and (36), we can get $\hat{p}_1 = \sqrt{3}$, $\hat{w}_1 = \frac{2}{3}$, $\hat{w}_2 = \frac{1}{6}$. Note that this set of univariate quadrature points and weights are the same as the set of univariate GHQ points and weights ($m_q = 3$). Because the univariate quadrature point sets with different accuracy levels satisfy the condition of Proposition 3.2, we only need to use N_{L-1}^n to calculate these points. Note that $L = 3$ and thus the largest q in N_q^n is 2 and we get the expression in Box III.

The accuracy level sequence $(1, \dots, 3, \dots, 1)$ generates 2 new points $[0, \dots, \pm\hat{p}_1, \dots, 0]^T$ and one old point $[0, \dots, 0, \dots, 0]^T$. There are C_n^1 kinds of such accuracy level sequences with one element being 3 and others being 1. The accuracy level sequence $(1, \dots, 2, \dots, 2, \dots, 1)$ generates 4 new points $([0, \dots, \pm\hat{p}_1, \dots, \hat{p}_1, \dots, 0]^T, [0, \dots, \pm\hat{p}_1, \dots, -\hat{p}_1, \dots, 0]^T)$ and one old point $[0, \dots, 0, \dots, 0, \dots, 0]^T$. There are C_n^2 kinds of such accuracy level sequences with two elements being 2 and others being 1.

Therefore, the total number of SGQ points is

$$4C_n^2 + 2C_n^1 + 1 = 2n^2 + 1.$$

After some algebraic operations, the following weights and points can be found (see Box IV).

Note that $\gamma_i = [0, \dots, \sqrt{3}, \dots, -\sqrt{3}, \dots, 0]^T$ denotes the point set in which $\sqrt{3}$ and $-\sqrt{3}$ appear in different places with $\sqrt{3}$ before $-\sqrt{3}$, and thus there are C_n^2 such different SGQ points. Other γ_i follow the same notation.

Case 2: $\hat{p}_1 = \hat{p}_2 \neq \hat{p}_3$

In this case, the point set $\{-\hat{p}_1, 0, \hat{p}_1\}$ with the weights $(\hat{w}_2, \hat{w}_1, \hat{w}_2)$ is used for level-2 and the point set $\{-\hat{p}_3, -\hat{p}_1, 0, \hat{p}_1, \hat{p}_3\}$ with the weights $(\hat{w}_5, \hat{w}_4, \hat{w}_3, \hat{w}_4, \hat{w}_5)$ is used for level-3. There are seven variables, $\hat{p}_1$, $\hat{p}_3$, $\hat{w}_1, \hat{w}_2, \hat{w}_3, \hat{w}_4$, and $\hat{w}_5$, where $\hat{p}_1$ and $\hat{p}_3$ are two tunable parameters. The weights $\hat{w}_i$ ($i = 1, \dots, 5$) are calculated from Eqs. (35) and (37).

Similarly with Case 1, the accuracy level sequence $(1, \dots, 3, \dots, 1)$ in N_2^n generates 4 new points $([0, \dots, \pm\hat{p}_1, \dots, 0]^T, [0, \dots, \pm\hat{p}_3, \dots, 0]^T)$ and one old point $[0, \dots, 0, \dots, 0]^T$. There are C_n^1 kinds of such accuracy level sequences with one element being 3 and others being 1. The accuracy level sequence $(1, \dots, 2, \dots, 2, \dots, 1)$ generates 4 new points $([0, \dots, \pm\hat{p}_1, \dots, \hat{p}_1, \dots, 0]^T, [0, \dots, \pm\hat{p}_1, \dots, -\hat{p}_1, \dots, 0]^T)$ and one old point $[0, \dots, 0, \dots, 0, \dots, 0]^T$. There are C_n^2 kinds of such accuracy level sequences with two elements being 2 and others being 1. Thus, the total number of SGQ points is

$$4C_n^1 + 4C_n^2 + 1 = 2n^2 + 2n + 1.$$

The weights (calculated from Algorithm 1) and points can be obtained by using Eqs. (49) and (50) given in Box V.

$$\mathbf{N}_{L-1}^n = \mathbf{N}_2^n = \left\{ \underbrace{(3, 1, \dots, 1), \dots, (1, \dots, 3, \dots, 1), \dots, (1, 1, \dots, 3),}_{C_n^1 \text{ elements}} \right. \\ \left. \underbrace{(2, 2, \dots, 1), \dots, (1, \dots, 2, \dots, 2, \dots, 1), \dots, (1, \dots, 2, 2)}_{C_n^2 \text{ elements}} \right\}.$$

Box III.

$$W_i = \begin{cases} \frac{n^2}{18} - \frac{7n}{18} + 1; & i = 1 \\ -\frac{n}{18} + \frac{2}{9}; & i = 2, \dots, 2n+1 \\ \frac{1}{36}; & i = 2n+2, \dots, 2n^2+1 \end{cases} \quad (47)$$

$$\gamma_i = \begin{cases} [0, \dots, 0]^T; & i = 1 \\ [0, \dots, \sqrt{3}, \dots, 0]^T; & i = 2, \dots, n+1; C_n^1 \text{ points} \\ [0, \dots, -\sqrt{3}, \dots, 0]^T; & i = n+2, \dots, 2n+1; C_n^1 \text{ points} \\ [0, \dots, \sqrt{3}, \dots, \sqrt{3}, \dots, 0]^T; & i = 2n+2, \dots, \frac{n(n-1)}{2} + 2n+1; C_n^2 \text{ points} \\ [0, \dots, -\sqrt{3}, \dots, -\sqrt{3}, \dots, 0]^T; & i = \frac{n(n-1)}{2} + 2n+2, \dots, n(n-1) + 2n+1; C_n^2 \text{ points} \\ [0, \dots, \sqrt{3}, \dots, -\sqrt{3}, \dots, 0]^T; & i = n(n-1) + 2n+2, \dots, \frac{3n(n-1)}{2} + 2n+1; C_n^2 \text{ points} \\ [0, \dots, -\sqrt{3}, \dots, \sqrt{3}, \dots, 0]^T; & i = \frac{3n(n-1)}{2} + 2n+2, \dots, 2n^2+1; C_n^2 \text{ points.} \end{cases} \quad (48)$$

Box IV.

$$W_i = \begin{cases} \frac{1}{2}(n-1)(n-2+n\hat{w}_1^2) - n(n-1)\hat{w}_1 + n\hat{w}_3; & i = 1 \\ (n-1)\hat{w}_2(\hat{w}_1-1) + \hat{w}_4; & i = 2, \dots, 2n+1 \\ \hat{w}_5; & i = 2n+2, \dots, 4n+1 \\ \hat{w}_2^2; & i = 4n+2, \dots, 2n^2+2n+1 \end{cases} \quad (49)$$

$$\gamma_i = \begin{cases} [0, \dots, 0]^T; & i = 1 \\ [0, \dots, \hat{p}_1, \dots, 0]^T; & i = 2, \dots, n+1; C_n^1 \text{ points} \\ [0, \dots, -\hat{p}_1, \dots, 0]^T; & i = n+2, \dots, 2n+1; C_n^1 \text{ points} \\ [0, \dots, \hat{p}_3, \dots, 0]^T; & i = 2n+2, \dots, 3n+1; C_n^1 \text{ points} \\ [0, \dots, -\hat{p}_3, \dots, 0]^T; & i = 3n+2, \dots, 4n+1; C_n^1 \text{ points} \\ [0, \dots, \hat{p}_1, \dots, \hat{p}_1, \dots, 0]^T; & i = 4n+2, \dots, \frac{n(n-1)}{2} + 4n+1; C_n^2 \text{ points} \\ [0, \dots, -\hat{p}_1, \dots, -\hat{p}_1, \dots, 0]^T; & i = \frac{n(n-1)}{2} + 4n+2, \dots, n(n-1) + 4n+1; C_n^2 \text{ points} \\ [0, \dots, \hat{p}_1, \dots, -\hat{p}_1, \dots, 0]^T; & i = n(n-1) + 4n+2, \dots, \frac{3n(n-1)}{2} + 4n+1; C_n^2 \text{ points} \\ [0, \dots, -\hat{p}_1, \dots, \hat{p}_1, \dots, 0]^T; & i = \frac{3n(n-1)}{2} + 4n+2, \dots, 2n^2+2n+1; C_n^2 \text{ points.} \end{cases} \quad (50)$$

Box V.

Case 1 is a special case of Case 2. If $\hat{p}_1 = \hat{p}_2 = \hat{p}_3 = \sqrt{3}$, one can get $\hat{w}_1 = \hat{w}_3 = \frac{2}{3}$, $\hat{w}_2 = 2\hat{w}_4 = 2\hat{w}_5 = \frac{1}{6}$. Additionally $[0, \dots, \pm\hat{p}_1, \dots, 0]^T$ and $[0, \dots, \pm\hat{p}_3, \dots, 0]^T$ become identical and the weights become

$$\frac{1}{2}(n-1)(n-2+n\hat{w}_1^2) - n(n-1)\hat{w}_1 + n\hat{w}_3 \\ = \frac{n^2}{18} - \frac{7n}{18} + 1, \\ (n-1)\hat{w}_2(\hat{w}_1-1) + \hat{w}_4 + \hat{w}_5 = -\frac{n}{18} + \frac{2}{9}, \quad \text{and} \quad \hat{w}_2^2 = \frac{1}{36},$$

which are the same weights as those in Case 1.

Case 3: $\hat{p}_1 \neq \hat{p}_2 \neq \hat{p}_3$

In this case, the point set $\{-\hat{p}_1, 0, \hat{p}_1\}$ with the weights $(\hat{w}_2, \hat{w}_1, \hat{w}_2)$ is used for level-2 and the point set $\{-\hat{p}_3, -\hat{p}_2, 0, \hat{p}_2, \hat{p}_3\}$ with the weights $(\hat{w}_5, \hat{w}_4, \hat{w}_3, \hat{w}_4, \hat{w}_5)$ is used for level-3. There are eight variables, $\hat{p}_1, \hat{p}_2, \hat{p}_3, \hat{w}_1, \hat{w}_2, \hat{w}_3, \hat{w}_4$, and $\hat{w}_5$, where $\hat{p}_1, \hat{p}_2, \hat{p}_3$ are three tunable parameters. The weights $\hat{w}_i (i = 1, \dots, 5)$ are calculated from Eqs. (35) and (37).

Similarly with Case 1, the accuracy level sequence $(1, \dots, 3, \dots, 1)$ contains 4 new points $([0, \dots, \pm\hat{p}_2, \dots, 0]^T, [0, \dots, \pm\hat{p}_3, \dots, 0]^T)$ and one old point $[0, \dots, 0, \dots, 0]^T$. There are C_n^1

$$W_i = \begin{cases} \frac{1}{2} (n-1) (n-2 + n\hat{w}_1^2) - n(n-1) \hat{w}_1 + n\hat{w}_3; & i = 1 \\ (n-1) \hat{w}_2 (\hat{w}_1 - 1); & i = 2, \dots, 2n+1 \\ \hat{w}_4; & i = 2n+2, \dots, 4n+1 \\ \hat{w}_5; & i = 4n+2, \dots, 6n+1 \\ \hat{w}_2^2; & i = 6n+2, \dots, 2n^2 + 4n+1 \end{cases} \quad (51)$$

$$y_i = \begin{cases} [0, \dots, 0]^T; & i = 1 \\ [0, \dots, \hat{p}_1, \dots, 0]^T; & i = 2, \dots, n+1; C_n^1 \text{ points} \\ [0, \dots, -\hat{p}_1, \dots, 0]^T; & i = n+2, \dots, 2n+1; C_n^1 \text{ points} \\ [0, \dots, \hat{p}_2, \dots, 0]^T; & i = 2n+2, \dots, 3n+1; C_n^1 \text{ points} \\ [0, \dots, -\hat{p}_2, \dots, 0]^T; & i = 3n+2, \dots, 4n+1; C_n^1 \text{ points} \\ [0, \dots, \hat{p}_3, \dots, 0]^T; & i = 4n+2, \dots, 5n+1; C_n^1 \text{ points} \\ [0, \dots, -\hat{p}_3, \dots, 0]^T; & i = 5n+2, \dots, 6n+1; C_n^1 \text{ points} \\ [0, \dots, \hat{p}_1, \dots, \hat{p}_1, \dots, 0]^T; & i = 6n+2, \dots, \frac{n(n-1)}{2} + 6n+1; C_n^2 \text{ points} \\ [0, \dots, -\hat{p}_1, \dots, -\hat{p}_1, \dots, 0]^T; & i = \frac{n(n-1)}{2} + 6n+2, \dots, n(n-1) + 6n+1; C_n^2 \text{ points} \\ [0, \dots, \hat{p}_1, \dots, -\hat{p}_1, \dots, 0]^T; & i = n(n-1) + 6n+2, \dots, \frac{3n(n-1)}{2} + 6n+1; C_n^2 \text{ points} \\ [0, \dots, -\hat{p}_1, \dots, \hat{p}_1, \dots, 0]^T; & i = \frac{3n(n-1)}{2} + 6n+2, \dots, 2n^2 + 4n+1; C_n^2 \text{ points.} \end{cases} \quad (52)$$

Box VI.

kinds of such accuracy level sequences with one element being 3 and others being 1. The accuracy level sequence $(1, \dots, 2, \dots, 2, \dots, 1)$ generates 4 new points not on the axes $([0, \dots, \pm\hat{p}_1, \dots, \hat{p}_1, \dots, 0]^T, [0, \dots, \pm\hat{p}_1, \dots, -\hat{p}_1, \dots, 0]^T)$, 4 new points located on the axes $([0, \dots, \pm\hat{p}_1, \dots, 0, \dots, 0]^T, [0, \dots, 0, \dots, \pm\hat{p}_1, \dots, 0]^T)$, and one old point $[0, \dots, 0, \dots, 0, \dots, 0]^T$. There are C_n^2 kinds of such accuracy level sequences with two elements being 2 and others being 1. However, since the points on the axes $([0, \dots, \pm\hat{p}_1, \dots, 0, \dots, 0]^T, [0, \dots, 0, \dots, \pm\hat{p}_1, \dots, 0]^T)$ can be generated by other sequences in $\mathbf{N}_n^2$, e.g. $(1, \dots, 2, \dots, 1, \dots, 2)$ and $(2, \dots, 1, \dots, 2, \dots, 1)$, there are $2C_n^1$ such points on the axes considering these duplicate points. So the total number of SGQ points is

$$4C_n^1 + 4C_n^2 + 2C_n^1 + 1 = 2n^2 + 4n + 1.$$

The weights (calculated from Algorithm 1) and the points can be obtained by using Eqs. (51) and (52) given in Box VI:

Case 2 is a special case of Case 3. If $\hat{p}_1 = \hat{p}_2$, then $[0, \dots, \pm\hat{p}_1, \dots, 0]^T$ and $[0, \dots, \pm\hat{p}_2, \dots, 0]^T$ become identical and the weights for these redundant points should be re-calculated as $(n-1)\hat{w}_2(\hat{w}_1-1) + \hat{w}_4$, the same as the corresponding weights in Case 2.

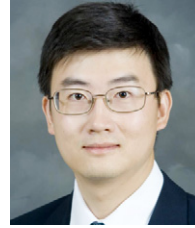
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